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DeepPipeNet enables deep learning based anomaly detection for oil and gas pipeline monitoring

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Abstract

Oil and gas pipelines are critical infrastructures that facilitate the transportation of energy resources over vast distances. However, they are prone to various operational threats, including leaks, corrosion, and mechanical failures, which can lead to severe environmental damage and financial losses. Traditional monitoring systems often struggle with real-time anomaly detection, necessitating advanced AI-driven solutions for improved pipeline integrity and risk mitigation. This study introduces DeepPipeNet, a hybrid deep learning-based ensemble framework designed to detect pipeline anomalies and failures with high precision. The proposed methodology integrates three state-of-the-art Convolutional Neural Networks (CNNs)—VGG16, ResNet50, and DenseNet121—to extract rich and diverse feature representations. These features are fused through a concatenation mechanism, followed by an attention module to emphasize critical patterns, and finally classified via a Meta-CNN architecture consisting of multi-path dense layers with softmax activation. DeepPipeNet was rigorously evaluated on two domain-specific datasets, the Oil Pipeline Accidents Dataset (focusing on multi-category cause classification) and Oil and Gas Pipline Leakage dataset focused on Corrosion Severity (categorized into high, medium, and low severity levels). After rigorous hyperparameter tuning to optimize generalization, the framework achieved outstanding test accuracy of 98.29% and 98.51%, respectively. These results demonstrate DeepPipeNet's superior capability in detecting pipeline-related anomalies with high precision (exceeding 98%), significantly minimizing false positives and enabling near-real-time monitoring. By leveraging deep feature fusion and attention-driven refinement, DeepPipeNet offers a scalable and robust AI-based solution that advances predictive maintenance strategies and ensures safer, more efficient pipeline operations in the oil and gas industry.

Keywords Deep learning, Pipeline anomaly detection, Oil and gas pipelines, DeepPipeNet, SMOTE, Multi-model feature fusion, VGG16, ResNet50, DenseNet, Feature concatenation, Attention mechanism, Channel attention, Spatial attention, Meta-CNN classifier, Adam optimizer, Corrosion severity prediction, Accident risk classification

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1 Introduction

Oil and gas pipelines are essential infrastructures that pass vital energies that cover an extensive geographical region. However, they are also likely to suffer operational challenges such as leakage, corrosion and mechanical breakdowns, which may have disastrous environmental and financial consequences It is imperative to note that the integrity of the pipeline must be monitored efficiently and in real-time to ensure safe working and less risks. Traditional monitoring tools, primarily sensor-based and manual and check resources, are, as a rule, inadequate to detect anomalies in due time and cannot provide valuable [1], actionable information in real-time.

Existing anomaly detectors, such as rule-based systems, statistical models, and traditional machine learning approaches, are usually not very satisfactory to handle the complexity of pipeline surveillance. These techniques are susceptible to flaws such as late detection delays, high levels of false positives and low levels of detecting new or unforeseen issues [2]. Moreover, traditional models are generally associated with substantial volumes of manual feature design and deal with large volumes of sensor data. Thus, there is an urgent necessity to find complex AI-powered solutions that can automatically identify anomalies in the pipes and provide precise real-time results.

The field of deep learning has demonstrated significant promise with respect to these challenges [3–5], where the power to learn intricate feature representations on big data without human intervention is achieved. In particular, Convolutional Neural Networks (CNNs) have been found useful in many anomaly detection tasks, such as in the detection of faults in a given industrial system. However, high false positives [6], low interpretability, and also have limited capabilities to perform with multi-dimensional and dynamic-time pipeline data remain a challenge to most of the existing deep learning solutions.

To address the shortcomings we introduce DeepPipeNet, a heterogeneous deep learning network that is specifically adapted to the task of anomaly detection in oil and gas pipelines. DeepPipeNet takes advantage of the strengths of three popular CNN architectures, including VGG16, ResNet50 and DenseNet121, to train various sets of characteristics based on the pipeline data readings. Each of these models is carefully chosen to bring out deep hierarchical properties (VGG16), make the most of residual learning to enable better gradient propagation through the network (ResNet50), and then build tight connection among features (DenseNet121) [7]. The extracted features are then fused together using a concatenation mechanism so as to combine the features and high density representation of the input data can happen.

Besides enhancing the model even more, the DeepPipeNet model also incorporates an attention module, which focuses on critical channel and spatial features, which improves the ability of the model to find subtle and complex anomalies [8]. Finally, the classification with high accuracy is carried out with the help of a Meta-CNN module constituted of dense layers and softmax activation that identifies the causes of anomalies and corrosion degree.

The primary objectives of our research are as mentioned below:

- To enhance anomaly detection in oil and gas pipelines by leveraging high-quality datasets such as the Oil Pipeline Accidents Dataset and the Oil and Gas Pipeline Leakage Dataset, focusing on capturing critical anomaly patterns and implementing

data preprocessing techniques, including feature normalization and dimensionality reduction, to improve model efficiency and reliability.

- To develop a hybrid deep learning-based framework (DeepPipeNet) that integrates VGG16, ResNet50, and DenseNet121 CNN architectures to effectively extract deep hierarchical, residual, and dense feature representations from pipeline sensor data, thereby enhancing anomaly detection accuracy in high-dimensional and dynamic environments.
- To improve anomaly detection in pipeline systems by employing advanced feature fusion techniques and an attention module, ensuring the framework's adaptability to various anomaly types such as leaks, corrosion, and mechanical failures, while reducing false positives and improving precision
- To rigorously evaluate DeepPipeNet's performance using performance metrics such as accuracy, precision, recall, and F1-score, benchmarking it against state-of-the-art pipeline monitoring systems to demonstrate its effectiveness in detecting and classifying pipeline-related anomalies.

DeepPipeNet is an enhancement of real-time oil and gas pipeline anomaly detection through the combination of hybrid deep learning structures, high-level feature fusion, and attention mechanisms. Not only does this technique enhance the measure of a wide range of anomalies associated with the pipeline, but also it performs well in high-dimensional and dynamic environments. The ability of the system to accurately classify pipeline issues such as leakages, corrosion and mechanical malfunction ensures great benefits of predictive maintenance and reduction of hazards. This study allows the entry into more resistant, scaled, and economical pipeline monitoring mechanisms that can reduce the risk of disastrous failure to the utmost, allowing oil and gas enterprises to operate in a safer and more sustainable way.

The remaining sections of this paper are structured as follows: Sect. 2 comprises of literature review, related studies, and techniques of anomaly detection in industrial pipeline systems. Section 3 describes the proposed methodology, such as the description of datasets, the data preprocessing procedures, the model structure, and evaluation metrics. Section 4 gives the experimental findings, with DeepPipeNet performance being highlighted. Lastly, the paper introductions its final Sect. 5, which contains conclusions and future research priorities, including real-time monitoring integration, and innovative predictive maintenance plans of pipeline operations.

2 Literature review

The growing reliance on complex technologies that identify the leaks in the pipeline has caused significant substantial advances in machine learning, deep learning, and other hybrid methods to improve the safety and efficacy of the oil and gas producing plants. As the pipeline systems and conditions of their operation grew dynamic, traditional detection tools, i.e., the statistical and the rule-based ones, could not address the requirements of the cryptic and dynamic leaks as well as the heterogeneity of the environmental conditions. Machine learning models, particularly Convolutional Neural Network (CNNs) and hybrid-based systems have been discovered to be useful to detect leakage during real-time with better accuracy and lower false positive rates. Despite such advances, challenges such as noise interference, real time processing of data and scalability still exist and a need to explore systems that are more resilient and flexible

is presented [9]. This review is a summary of the key results of recent studies and how deep learning and sensor technologies can be used together to create more robust, real-time pipeline monitoring systems that can be used to its benefit to mitigate the risks and make operations safer.

2.1 Machine learning approaches for leak detection

Machine learning has been applied widely to enhance the accuracy of the pipeline leak recognition capabilities offering mighty instruments to conduct the parameter and sensor data analysis. Ekong et al. [10] developed a system that integrated Convolutional Neural Networks (CNNs), and Internet of Things (IoT) to detect leaks of oil pipes with 97% accuracy. The combination provided the ability to monitor pipeline leaks in time, improving the safety of the pipelines. Aljameel et al. employed machine learning methods such as Support Vector Machines (SVM), Random Forest, and Decision Trees to detect pipeline leakages. Among the models with an accuracy rate of 97.4 percent SVM was the best. Their results demonstrated that SVM was able to detect anomalies in pipelines, but it was hard to scale to a real-life application.

Ihsan et al. [11] applied the Artificial Neural Networks (ANNs) in the detection of the oil pipeline leakages by predicting the position of the leakages using the outlets pressure and leak rates. They also addressed the issues of lack of data by creating simulated data of leak processes which provides a potent solution in improving the safety of pipelines. In Wang et al. machine learning and Distributed Acoustic Sensing (DAS) technology were incorporated in managing pipeline integrity. The integration led to much improvement on leak detection and real time monitoring of surroundings of the pipes and improved the accuracy of event detection and classification. Chang et al. [12] suggested an AI platform of oil–gas pipeline metering and leakage alarm, which incorporated a few sensors and circuits to monitor and determine the leaks in pipeline in real-time. The system enhanced monitoring and control which offered greater reliability and safety where real time alarms and preventive measures were offered.

Yadukrishnan et al. [13] suggest a gas pipeline leak detection technique which is based on acoustic signal analysis with accuracy of 76 percent with the help of the Random Forest classifier. Another study of this nature that also explored the efficacy of acoustic signal analysis in leak detection was conducted by Ullah et al. [14] investigated acoustic emission (AE) technology with machine learning to obtain a remarkable 99% classification accuracy in leak detection. This was very effective in the detection of pinhole-sized leaks under various conditions, giving testament to the strength of AE technology in real-world applications for pipeline health monitoring. Chhotaray and Kulshreshtha [15] used machine learning algorithms on sensor readings from pipeline inspection gauges (PIG) sensors, resulting in a drastic reduction in false alarms. Their method, which eliminated more than 90% of false positives, greatly improved the accuracy of defect detection, providing more accurate pipeline health monitoring.

2.2 Advanced deep learning models for leak detection

The methods of deep learning, particularly CNNs, are becoming increasingly popular due to their ability to process high-dimensional data with complex features and learn to achieve good results by automatically extracting meaningful features. Xu et al. proposed the SART-1DCNN model which detected the presence of leaks in oil pipes after

considering leakage signals as well as noise data and advanced tools including attention mechanisms and residual blocks with an impressive 99.44 percent accuracy. AlAzri et al. [16] have used Pyramid Scene Parsing Network (PSPNet) to detect the presence of leaks in oil pipes using drone images. This is a deep learning approach, which uses U-Net segmentation, to enhance leaks detectability, which led to substantial cost savings based on early warnings.

Yang et al. [17] suggested a signal denoising algorithm (VMD-MD) using a 1D CNNs in the detection of pipeline leakage. The hybrid model also improved the accuracy of leak detection, because any interference of the noises was eliminated, which proved it to be effective in the course of pipeline operations under various conditions. In Zhang et al. [18] the authors introduced an AI-based metering and leakage alarming platform of oil–gas pipes by introducing such multiple sensors and circuits to detect and monitor pipeline leakage in real-time. The system also enhanced the monitoring capacity and control hence ensuring greater reliability and safety through real time alarming and preventive measures. Avcili and Kartal [19] used CNNs on the leaks of underwater pipes with a 97.63 per cent accuracy. This paper has revealed the ability of deep models in detecting ecological danger as a result of pipeline breakage.

2.3 Hybrid approaches and IoT integration

The combination of deep learning and IoT has led to new solutions of real-time expert leakage in the pipeline. Mehta and Manisha [20] proposed a monitoring system that is based on CNNs and IoT cameras that continuously process a video image in order to detect pipeline leak, blockage, and corrosion. Their model had greatly enhanced the health management of the pipeline because it allowed a continuous and accurate monitoring of the pipeline conditions. Similarly, Wang and Lin [21] designed an AI-driven pipeline leakage detector in gas pipelines, and used CNNs to analyze pressure variation and detect gas leakages at large distances. It has been tested in reality, and its performance proved that it is possible to monitor pipeline in a large scale.

A survey of various signal processing and AI-based leak detectors was done by Rajasekaran and Kothandaram [22] They found that the hybrid approach relating to signal processing with AI models like 1D CNNs was the most accurate in leaks detection (98.32), and this may be explained by the importance of hybrid approaches in improving the efficiency of real-time detection. Alharam et al. [23] explored the use of drones bearing thermal cameras and on-board processing based on the artificial intelligence to inspect leaks and cracks in pipes. A real-time detection system with a delay of less than 100 ms as demonstrated was shown to enhance the pipe safety as well as save plenty of money and time as far as inspection is concerned.

2.4 Infrared and image-based detection techniques

Image-based and infrared detection methods have been useful in non-invasive pipeline infrastructure monitoring. Obaid and Hamad [24] proposed a drone system for oil pipeline leak detection based on deep learning techniques for edge detection and contour recognition. Their application of the DexiNed algorithm, which effectively identifies oil slicks, highlighted the capability of aerial monitoring and deep learning in pipeline surveillance. Nuthalapati et al. [25] also suggested a deep learning-based gas pipeline leak detection system with the help of infrared cameras. The system had a 98.7% prediction

ratio and 97.4% classification accuracy, proving its better performance in real-time detection of methane gas leaks.

Kyriacou et al. [26] also developed a model that integrates machine learning, data assimilation, and physics of fluid flows to detect the probability of having leaked pipes in real-time. This technique reduced leak response time and fugitive emissions that demonstrated tremendous enhancement in leak detection and response. Cui et al. [27] proposed a better pipeline leakage detection algorithm, which was developed by utilizing ICEEMDAN signal processing algorithm and probabilistic neural networks (PNN). The model was 98 percent accurate even with the interference of noise, as well, which demonstrated the usefulness of hybrid methods in unfavorable working environments. Another study proposed by Gao et al. [28] involved the use of a Particle Swarm Optimization (PSO)-optimized Support Vector Machine (SVM) algorithm to detect pipeline leaks that was better than the traditional models. This hybrid approach enhanced the leak detection performance in complex pipeline systems by doing wavelet domain denoising and extracting important features of the signal.

Based on the comprehensive review of existing approaches as shown in the Table 1, several critical research gaps in oil and gas pipline anomalies can be identified:

- *Limited scalability and generalization* Most models, including those by Ekong et al. [29] and Aljameel et al. [1], are highly accurate in controlled conditions but are not scalable to dynamic real-world pipeline conditions. The use of IoT or particular sensors limits their usage to varied environments.
- *Hybrid system requirements* Various works, such as Ihsan et al. [2] and Rajasekaran and Kothandaraman [13], point to the versatility of hybrid methods but continue to center largely around individual methodologies. Merging machine learning with advanced signal processing, IoT, or deep learning technologies can potentially provide increased robustness and flexibility to tackle sophisticated, multi-factor leak situations.
- *Real-time processing limitations* Programs such as Xu et al. [8] and Wang et al. [3] necessitate enormous processing capabilities and delay real-time processing, especially with big pipe systems. Systems that come after this have to put effort into keeping the latency lower and precision consistent, notably on leak detection before the onset.
- *Environmental sensitivity and noise* Certain models like Ullah et al. [6] and Yadukrishnan et al. [5] use acoustic signals and other environmental inputs which are vulnerable to noise and external factors. More robust systems that can distinguish between noise and real leaks should be developed.
- *Cost and technological dependence* Drones and infrared-based technologies, which were discussed by Obaid and Hamad [15] and Nuthalapati et al. [30], have good performance but depend significantly on expensive technology and meteorological conditions. Practical and economical large-scale implementation requires solutions with less dependence on expensive hardware.

Our suggested hybrid DeepPipeNet solution solves these research limitations in the following main innovations:

Table 1 Comparative analysis of related works in pipeline leak detection

Study	Technique/algorithm	Focus Area	Pros	Cons	Research Gap
Ekong et al. [29]	Convolutional Neural Networks (CNNs) + IoT	Oil pipeline leak detection	Real-time monitoring High detection accuracy	Limited scalability Relies on IoT integration	Lacks scalability in diverse operational conditions
Aljameel et al. [1]	Support Vector Machines (SVM), Random Forest, Decision Trees	Multi-algorithm comparison	SVM outperforms others Effective for pipeline anomalies	Scalability challenges Limited to certain models	Need for hybrid approach for enhanced robustness
Ihsan et al. [2]	Artificial Neural Networks (ANNs)	Leak position forecasting	Addresses data scarcity Simulated data generation	Simulated data might not fully represent real leaks	Lack of real-time leak detection integration
Wang et al. [3]	Machine learning + Distributed Acoustic Sensing (DAS)	Pipeline integrity management	Real-time monitoring Increases leak detection accuracy	High sensor dependency Requires advanced sensors	Requires enhanced real-time data processing capabilities
Yadukrishnan et al. [5]	Acoustic Signal Analysis + Random Forest	Gas pipeline leak detection	Effective for real-time monitoring Low-cost solution	Lower accuracy Relies on acoustic signals	Needs improvements in detection accuracy
Ullah et al. [6]	Acoustic Emission (AE) + Machine Learning	Pinhole leak detection	Highly effective for small leaks Real-world applicability	Sensitive to environmental conditions Expensive setup	Need for robustness in varying environmental conditions
Chhotaray and Kulshreshtha [7]	Machine Learning on PIG sensor data	Defect detection	Significant reduction in false alarms Improved accuracy	Still dependent on PIG sensors Limited to fixed systems	Lack of dynamic monitoring flexibility
Xu et al. [8]	SART-1DCNN (CNN-based)	Oil pipeline leak detection	High accuracy Effective noise handling	High computational requirements Needs high-quality data	Requires real-time, low-latency processing
AlAzri et al. [9]	Pyramid Scene Parsing Network (PSPNet) + U-Net	Drone-based oil pipeline leak detection	Effective for early leak prediction Image segmentation	High computational cost Requires drone systems	Needs optimization for cost-effectiveness
Rajasekaran and Kothandaraman [13]	Hybrid Signal Processing + 1D CNNs	Leak detection	High accuracy Hybrid approach improves detection	Limited to specific signal processing techniques	Could be improved by integrating more diverse methods
Alharam et al. [14]	Drones with Thermal Cameras + AI Onboard Processing	Leak and crack detection	Reduces inspection cost High real-time performance	Limited to thermal imagery Expensive technology	Lack of scalability and adaptability in various environments

Table 1 (continued)

Study	Technique/algorithm	Focus Area	Pros	Cons	Research Gap
Obaid and Hamad [15]	DexiNed Deep Learning + Drone Images	Edge detection for oil leaks	Effective at long distances Low-cost solution	Relies on drones Weather dependency	Requires further refinement in drone image processing
Nuthalapati et al. [30]	CNNs + Infrared Cameras	Gas pipeline leak detection	High prediction ratio Real-time detection	Limited to gas leaks Requires infrared cameras	Needs integration with other sensor modalities
Kyriacou et al. [16]	Machine learning + Data Assimilation + Fluid Flow Models	Real-time leak detection	Enhanced leak response time Probabilistic approa	Computationally intensive Needs precise fluid models	Lack of adaptability in real-time pipeline operations

- *Scalability and flexibility* Through the combination of several detection techniques, our system can scale to varied pipeline environments without sacrificing performance.
- *Real-time, low-latency processing* Tuned for low-latency processing, our solution is capable of processing real-time data streams for quick detection and response times.
- *Noise resistance* Sophisticated signal processing software assists in suppressing noise interference, enhancing the ruggedness of leak detection, especially in harsh conditions.
- *Cost-effective deployment* Utilizing IoT and cost-effective sensors, our solution reduces the need for expensive hardware while offering high accuracy and reliability.

3 Proposed methodology

The Proposed architecture is DeepPipeNet architecture based on deep learning and specifically in the oil and gas industry. It begins with the gathering of the pipeline data sets of normal operations and leakage conditions as illustrated in Fig. 1. The raw data is first introduced into a preprocessing stage in which strategies of noise reduction are used to enhance the quality of the signal and followed by the class balancing using SMOTE to address the issue of data imbalance. After data clean and normalization step, a multi-model feature extractor module is used based on three pre-trained convolutional neural networks VGG16, ResNet50 and DenseNet. The three models are used concurrently in order to retrieve robust and varied features that represent elaborate patterns in the data. The features obtained are subsequently concatenated and reduced (2048 to 512) dimensionally to ensure that the computations made are not too costly and relevant information is not lost.

This smaller feature set is inputted into a dual attention system that has both channel and spatial attention that concentrates the most important and informative features of the data. Attention enhanced features are subsequently forwarded to a Meta-CNN classifier which contains dense layers that are fully connected and dropout to regularize and finally softmax activation function to make final classification. The Adam optimizer with some hyper parameters is used to maximize the learning process. The model is trained to produce two outputs, one of which is predicting the severity of corrosion (scaled between 0 and 4) and the other being multi-classification of risk of accidents.

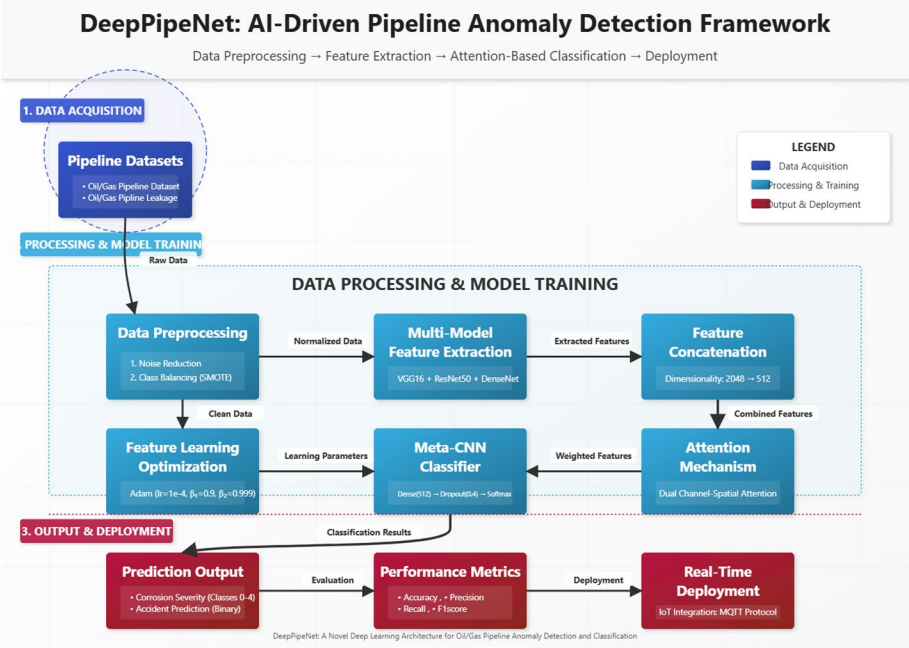


Fig. 1 Workflow of proposed methodology

Performance measures using performance metrics such as accuracy, precision, recall, and F1-score are used to measure the classification performance. Finally, the trained model is put into practice with the use of the IoT by means of the MQTT protocol in the real-time applications, which enables the detection of the anomaly in the real-time pipeline systems timely and effectively.

3.1 Dataset

We performed strict experiments to test the effectiveness of DeepPipeNet with reference to two domain-specific datasets: the Oil Pipeline Accidents Dataset (classifying the causes of the accident in multiple categories) and the Oil and Gas Pipeline Leakage Dataset (predicting the severity of corrosion). These datasets represent the actual operational threats to oil and gas pipelines such as leakages, corrosion, and mechanical failures, and allows us to greatly test our framework in real conditions. We elaborate on their structure below together with preprocessing and how it is relevant to the anomaly detection in pipelines.

3.1.1 Oil and gas pipeline leakage dataset

The Oil-and-Gas-Pipeline-Leakage Dataset was utilized in the research to study the oil and gas pipeline infrastructure corrosion defects with the purpose of categorizing the amount of corrosion into Low, Medium and High levels. The dataset was downloaded on an open-source dataset on GitHub https://github.com/Abimbola-ai/Oil-and-gas-pipeline-leakage [31] and has 10,292 instances of pipeline data, with eight numeric parameters each. The variables are critical such as wellhead pressure, wellhead temperature, gas production, oil production, water production, and size of corrosion defect among others, which are involved in ensuring the pipeline integrity and health.

The dataset is aimed to predict the target variable the corrosion defect (CR), which is utilized as a measure of the well-being of the pipeline. The designation of CR into three

categories (Low, medium, and high) enables researchers to quantify the extent of corrosion which is crucial in determining the appropriate maintenance and minimizing chances of pipeline failure. The data can be applied to regression and classification applications appropriately with the corrosion defects taking the first significance in terms of predicting the integrity of pipeline. In assessing the model, the data was separated into a test set and a training set. The training set will have 8,232 samples with the test set having 2,059 samples thus possible to have firm model assessment and performance metric measurement. The data was divided using train test split which would enable proper division of the data to be effectively validated.

Table 2 has lots of features in the data set, which include wellhead pressure and temperature, oil and gas production rate, elementary solid and water measurements, and gas gravity, among others. The features play a crucial role in creating predictive models of learning, and classification of the level of corrosion defects in pipes, hence the depth of information in predictive maintenance and risk management of oil and gas industry.

The target variable in this data set is the Corrosion Defect (CR) which is used to classify it. The CR values are put in three classes namely Low, Medium and High corrosion severity. Other features such as temperature, pressure, rate of gas flow etc. are numeric, and reflect the working conditions of the pipeline, which affect its overall integrity. These features are a prerequisite to predictive modeling that scales operators of the healthcare of pipes using these parameters. The database is designed in such a way that it provides practical information on the maintenance need of the pipelines with particular focus on corrosion flaws which constitute the main indicators of the possible breakdown of pipeline infrastructure.

The skewness of the three classes in the Oil-and-Gas-Pipeline-Leakage Dataset is illustrated in Fig. 2 and skewness in the distribution of the number of corrosion levels in the dataset should be considered when training Oil-and-Gas-Pipeline-Leakage Dataset models and measuring performance. The most frequent level of severity of 5,146 instances (50% of the dataset) is the High Corrosion, meaning that a significant portion of the pipeline infrastructure may be experiencing serious cases of corrosion. The level of degraded pipe of medium is 2,058 (20%) and represents the median level of corrosion, and the level of minimum corrosion of 3,089 (30%) represents the category of Low Corrosion. The given type of skewness of classes is quite typical in real-world data and justifies the necessity of managing such skewness when creating predictive models to prevent the bias of the model towards the more representative one (High Corrosion) under the usual conditions, in case of not addressing this skewness in the appropriate

Table 2 Data types of all features from oil-and-gas-pipeline-leakage dataset

Feature	Description	Data type
Wellhead temp. (°C)	Temperature measured at the wellhead	Float64
Wellhead press (psi)	Pressure recorded at the wellhead	Float64
MMCFD gas	Gas flow rate in million standard cubic feet per day	Float64
BOPD	Volume of oil produced per day, measured in barrels	Float64
BWPD	Volume of water produced per day, measured in barrels	Float64
BSW	The percentage of basic solids and water in the mixture	Float64
CO2 mol	Molecular mass of carbon dioxide present in the gas mixture	Float64
Gas Grav	The density of the gas compared to air	Float64
CR (Corrosion defect)	Severity level of corrosion in the pipeline (Low, Medium, High)	Categorical

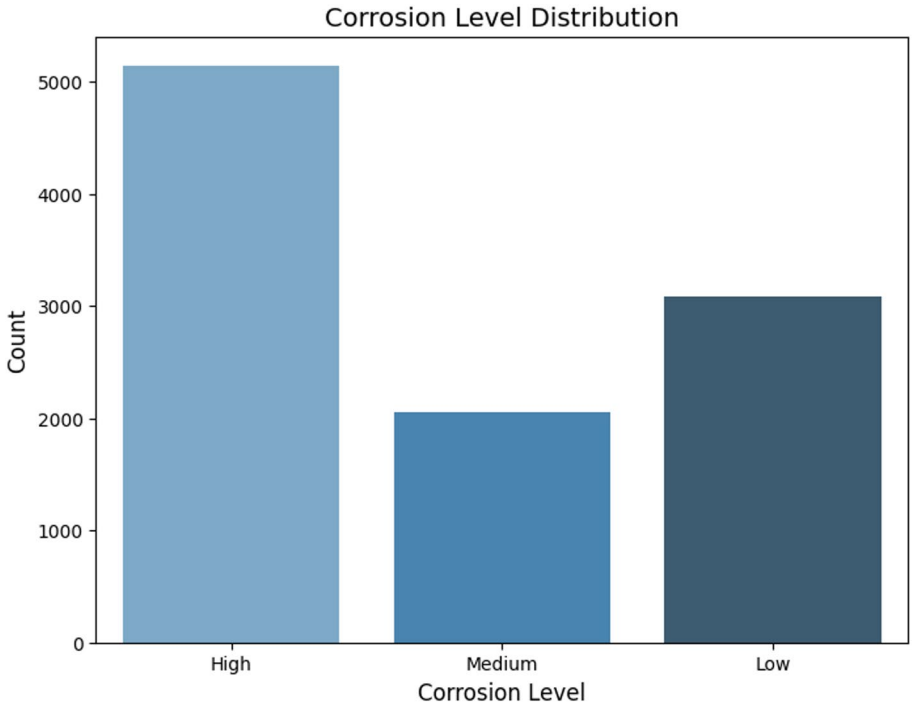


Fig. 2 Distribution of corrosion defect classes in oil and gas pipeline leakage dataset

way. This distribution should be identified to construct more effective and accurate predictive models to avoid pipeline failure and maintenance.

3.1.2 Oil pipeline accidents dataset

The Oil Pipeline Accidents Dataset is a sizeable dataset that can be accessed publicly, at the site https://www.kaggle.com/datasets/usdot/pipeline-accidents [32] and that was compiled in order to record the information about oil pipeline accidents, including their causes, environmental impact, and expense. The dataset consists of 2,795 samples and 48 attributes, which makes it a gold mine in insight and prediction of the pipeline accidents. It has such features as Accident Year, Accident Date/Time, Operator Name, and Pipeline Location that include such informative context to the pipeline system involved in every accident. We use this dataset in our work because we are interested in predicting the cause of pipeline accidents with the Cause Category, which is an important feature that determines the principal cause of the accident (e.g., equipment failure, external damage, etc.).

Vital information is also present in the dataset in terms of the effects of the accident as illustrated in Table 3, including the Release (Barrels), which comprises of Unintentional and Intentional releases of materials and Liquid Ignition and Liquid Explosions indicators. Moreover, the data on injuries and fatalities, including Operator Employee Injuries, Public Injuries, and All Fatalities are also a part of the dataset that should be used to evaluate the human implication of pipeline failures. Financial information, such as Property Damage Costs, Emergency Response Costs, as well as Environmental Remediation Costs is also recorded and the aggregate picture of the effect of the accident is provided.

In our study, we focus on the Cause Category feature and use it as the target variable to categorize pipeline accidents. It makes the data extremely helpful in training machine

Table 3 Data types of all features from oil pipeline accidents dataset

Feature	Description	Data type
Report number	Unique identifier for the report	Int64
Supplemental number	Additional identifier for multiple entries	Int64
Accident year	Year of the pipeline accident	Int64
Accident date/time	Date and time of the accident	Object
Operator ID	Unique ID of the operator	Int64
Operator name	Name of the pipeline operator	Object
Pipeline/facility name	Name of the pipeline or facility	Object
Pipeline location	Location details of the pipeline	Object
Pipeline type	Type of pipeline (e.g., transmission, gathering)	Object
Liquid type	Type of liquid involved in the incident	Object
Liquid subtype	Subtype of the liquid	Object
Liquid name	Specific name of the liquid	Object
Accident city	City where the accident occurred	Object
Accident county	County of the accident location	Object
Accident state	State of the accident location	Object
Accident latitude	Latitude coordinate of the accident site	Float64
Accident longitude	Longitude coordinate of the accident site	Float64
Cause category	Primary category of the cause	Object
Cause subcategory	Subcategory detailing the cause	Object
Unintentional release (barrels)	Volume of unintentional liquid release	Float64
Intentional release (barrels)	Volume of intentional liquid release	Float64
Liquid recovery (barrels)	Amount of liquid recovered	Float64
Net loss (barrels)	Net loss of liquid after recovery	Float64
Liquid ignition	Indicates if ignition occurred (Yes/No)	Object
Liquid explosion	Indicates if explosion occurred (Yes/No)	Object
Pipeline shutdown	Indicates if the pipeline was shut down	Object
Shutdown date/time	Date and time of shutdown	Object
Restart date/time	Date and time of pipeline restart	Object
Public evacuations	Number of public evacuations	Float64
Operator employee injuries	Injuries among operator employees	Float64
Operator contractor injuries	Injuries among contractor employees	Float64
Emergency responder injuries	Injuries among emergency responders	Float64
Other injuries	Other injuries not categorized	Float64
Public injuries	Injuries among the public	Float64
All injuries	Total number of injuries	Float64
Operator employee fatalities	Fatalities among operator employees	Float64
Operator contractor fatalities	Fatalities among contractor employees	Float64
Emergency responder fatalities	Fatalities among emergency responders	Float64
Other fatalities	Other fatalities not categorized	Float64
Public fatalities	Fatalities among the public	Float64
All fatalities	Total number of fatalities	Float64
Property damage costs	Cost of property damage	Float64
Lost commodity costs	Cost of lost commodity	Float64
Public/private property damage costs	Damage cost to public/private property	Float64
Emergency response costs	Cost of emergency response	Float64
Environmental remediation costs	Cost of environmental remediation	Float64
Other costs	Other miscellaneous costs	Float64
All costs	Total financial cost incurred	Int64

learning models to predict accidents, assess the risk, and increase the pipeline safety. The variety of qualities that are presented, such as practical specifics, economic and environmental impacts make this data an abundant source to study the phenomenon of pipeline accidents and create anticipatory mechanisms to reduce the number of such events in the future.

Since Fig. 3, reveals that the distribution in the cause category in the Oil Pipeline Accidents Dataset is skewed, as the various causes of pipeline accidents possess a high skewness with the most prevalent cause being Material/Weld/Equipment Failure. This type of category forms the largest part of the dataset, where the incidents are 1,435, which is the majority of the accidents. The second topicality is corrosion with 592 cases, which indicates the necessity to reduce pipeline aging and wear. The other important reasons include Incorrect Operation (378), and Natural Force Damage (118), namely, the part of the accident rate. The other classes like: Excavation Damage (97), All Other Causes (118), and Other Outside Force Damage (57) are small units of the data set. This skew in the classes suggests that the task needs models that are strong enough to handle skewed distributions of classes and give high-priority detection of some key safety causes like Material/Weld/Equipment Failure and Corrosion that are critical to the operations of the pipeline.

3.2 Input representation for CNN architectures

Both datasets can be perceived as tabular, and the data is characterized by the presence of 1D feature vectors. To exploit the spatial features that Convolutional Neural Networks can extract (like VGG16, ResNet50, and DenseNet121), a transformation from a 1D feature space to a 2D representation is mandatory. We adopted the matrix transformation approach.

One sample containing n features will reshape its 1D feature vector into a 2D matrix of size h x w, where h*w > = n. Zero-padding is applied to fill the matrix at certain positions as long as it satisfies the above condition. Hence, dimensions h and w are selected so that they form a compact grid which encourages effective application of convolutional filters onto it. For the Oil and Gas Pipeline Leakage Dataset, a vector of size 9 was padded to

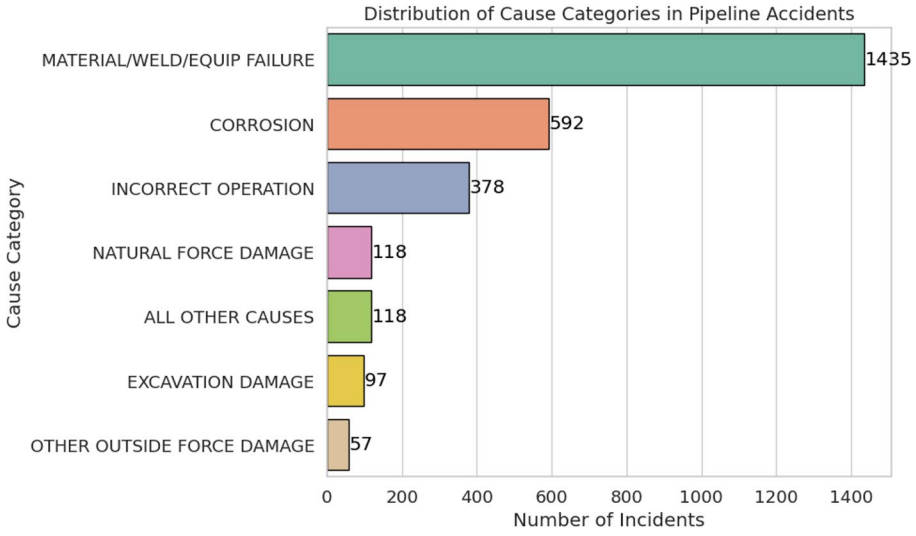


Fig. 3 Distribution of classes in oil pipeline accidents dataset

be reshaped into a 3 by 3 matrix (8 features per sample). The most important 36 features determined by a preliminary feature importance analysis using XGBoost were selected from the Oil Pipeline Accidents Dataset, and reshaped into a 6 by 6 matrix. The new formulation allows the CNNs to understand correlations between neighboring features in the 2D grid as spatial patterns, a method previously proven to perform well for datasets derived from tabular data [33].

3.3 Data pre-processing

DeepPipeNet is ensured to be reliable and generalizable by preprocessing the Oil and Gas Pipeline Leakage Dataset and Oil Pipeline Accidents Dataset. The major issues of imbalance between the classes, the values absent in the process and the outliers were dealt with in a systematic manner with the goal of maximizing the learning of features and the performance of models in dynamic pipeline settings.

Both Datasets there was Missing values, which when left without control, can bias feature distributions and cause a reduced model performance In the Data on oil pipeline accidents, numerical characteristics (e.g., Unintentional Release (Barrels) and Net Loss (Barrels)) had insignificant missingness (less than 2%), which was filled in by median values to preserve outlier sensitivity. Mode values were used to replace categorical attributes (e.g. Cause Category and Pipeline Type) in order to maintain uniformity. Similarly, in Pipeline Leakage Dataset, the features of Wellhead Pressure and Gas Flow rate had little missing data (less than 1.5 percent), which were processed with median values.

Interquartile Range (IQR) filtering was used to remove outliers which were typically a result of noise or some unusual conditions during the operations. Where the numeric features were involved (i.e., Wellhead Temperature and Gas Production), the value exceeding that of 1.5XIQR was clipped on the 5 th and 95 th percentiles. Distorted distributions among features, e.g. CO2 Mol. %, were transformed using the logarithmic operation that made the dependence on the size of outliers less sensitive. On the other hand, both datasets have evidence of widespread class skewness, posing the risk of model bias to dominant classes. The classes of the Pipeline Leakage Dataset Corrosion severity were not data-balanced: High (5,146, 50%), Low (3,089, 30%), and Medium (2,058, 20%). In order To address this, the Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively to the training set after the initial 80/20 train-test stratified split. The test set was kept completely separate and undisturbed to obtain a strict and realistic assessment of the model on the original, imbalanced data, simulating real-world deployment conditions. SMOTE synthesizes samples for the minority classes by interpolating between nearest neighbors in the training set feature space [34]. Post-processing resulted in a 250% increase in minority class representation in the training set as displayed in Fig. 4, while the test set maintained its original class distribution.

While SMOTE improves class balance and recall for minority classes, it also introduces synthetic samples that may not fully capture the complexity and variability of rare real-world anomalies. Because SMOTE generates new instances through interpolation, there is a risk of smoothing sharp decision boundaries or amplifying local feature correlations present in the training data. As a result, performance on extremely rare or previously unseen anomaly types in real operational environments may be lower than benchmark results suggest. To mitigate this risk, the test sets were left fully imbalanced, and model evaluation was performed exclusively on original, non-synthetic data. Future

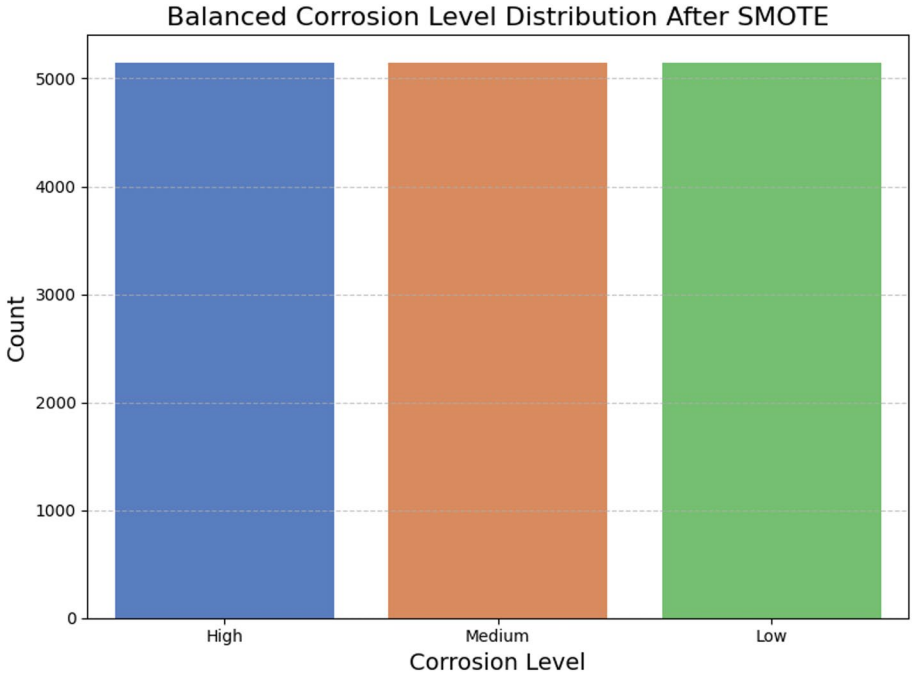


Fig. 4 Distribution of corrosion defect classes after SMOTE in oil and gas pipeline leakage dataset

work will explore alternative imbalance-handling strategies, including cost-sensitive learning and real anomaly augmentation from operational datasets.

For the Oil Pipeline Accidents Dataset, Cause Category distribution was strongly imbalanced, with Material/Weld/Equipment Failure being the most prominent (1,435, 51.3%), followed by Corrosion (592, 21.2%), and uncommon classes such as Natural Force Damage (118, 4.2%). SMOTE was selectively applied only to the training set for minority classes (Excavation Damage, Other Outside Force Damage, Natural Force Damage), increasing their representation by 400% as shown in Fig. 5, to avoid underfitting and increase detection of uncommon accident causes. The test set remained untouched with its original class distribution.

3.4 DeepPipeNet anomaly detection in oil and gas pipelines framework

In this section, we describe our hybrid DeepPipeNet framework specifically for oil and gas pipeline anomaly detection. DeepPipeNet overcomes the limitations of dynamic pipeline environments through multi-scale feature extraction, channel-spatial attention mechanisms, and adaptive meta-learning. The approach starts with preprocessing pipeline data, then employs three base CNNs—VGG16, ResNet50, and DenseNet121—to learn hierarchical, residual, and dense spatial patterns. VGG16 detects localized hotspots of corrosion, ResNet50 prevents gradient degradation for robust feature learning, and DenseNet121 retains fine-grained leakage signatures. Global Average Pooling (GAP) further compresses these features into compact vectors, lowering their dimensionality.

The concatenated feature vector $\boldsymbol{F}_{concat}$ undergoes channel-spatial attention weighting to prioritize critical anomalies (e.g., sudden pressure drops) while suppressing noise. The attention mechanism assigns learnable weights (W_c, W_s) to amplify discriminative features, enhancing the meta-learner's refinement capability. The meta-learner processes features through dense layers with dropout regularization, followed by softmax

classification to predict anomaly causes (e.g., corrosion, mechanical failure) or corrosion severity levels (low/medium/high).

Input: (Oil Pipeline Accidents and Pipeline Leakage Datasets) dataset D with N samples
Output: Anomaly cause classification or corrosion severity prediction, Performance metrics (Accuracy, Precision, Recall, F1-Score)
Procedure:

1. Data Pre-processing

- Remove duplicates: $D' = D \setminus \{ \text{duplicate entries} \}$
- Handle missing values: Replace $\emptyset$ with NaN , drop rows with NaN .
- Balance classes: Apply SMOTE to generate synthetic samples for minority classes.

Standardize features: $z = \frac{x-\mu}{\sigma} \forall x \in D'$.

2. Multi-Scale Spatial Feature Extraction:

2.1. Initialize Base CNNs:

- VGG16: Extracts hierarchical spatial patterns (e.g., corrosion hotspots).
- ResNet50: Leverages residual blocks to model high-dimensional dynamics (e.g., pressure, temperature).
- DenseNet121: Preserves dense connectivity for subtle leakage detection.
- VGG16: Extracts hierarchical spatial patterns (e.g., corrosion hotspots).

Extract Features:

$$F_{\text{vgg}} = \text{VGG16}(X), F_{\text{res}} = \text{ResNet50}(X), F_{\text{dense}} = \text{DenseNet121}(X)$$

2.2. Apply Global Average Pooling (GAP):

$$F_{\text{vgs-pooled}} = \text{GAP}(F_{\text{vgg}}), F_{\text{res-pooled}} = \text{GAP}(F_{\text{res}}), F_{\text{dense-pooled}} = \text{GAP}(F_{\text{dense}})$$

3. Channel-Spatial Attention Fusion:

3.1. Concatenate Features:

$$F_{\text{concat}} = F_{\text{vgs-pooled}} \oplus F_{\text{res-pooled}} \oplus F_{\text{dense-pooled}}$$

3.2. Compute Channel Attention Weights:

$$A_c = \sigma(W_c \cdot \text{GAP}(F_{\text{concat}}) + b_c)$$

3.3. Compute Spatial Attention Weights:

$$A_s = \sigma(W_s \cdot \text{Conv}(F_{\text{concat}}) + b_s)$$

3.4. Generate Attention-Weighted Features:

$$F_{\text{attn}} = A_c \odot (A_s \odot F_{\text{concat}})$$

4. Meta-Learner Refinement:

4.1. Flatten F_{atm} and process through dense layers:

$$F_{\text{dense1}} = \text{ReLU}(W_d \cdot F_{\text{attn}} + b_d)$$

4.2. Apply Dropout (0.4):

$$F_{\text{drop}} = \text{Dropout}(F_{\text{dense1}})$$

4.3. Softmax Classification:

$$\hat{y} = \text{softmax}(W_f \cdot F_{\text{drop}} + b_f)$$

5. Training Protocol:

- **Loss Function:** Weighted cross-entropy to mitigate class imbalance:

$$\mathcal{L} = -\sum_{i=1}^{C} w_i \cdot y_i \log(\hat{y}_i)$$

- where w_i is the inverse frequency of class i.
- Optimization: Adam ($\eta_0 = 0.001$, decay rate $= 0.01$) with L2 regularization ($\lambda = 0.01$).
- Regularization: Early stopping (patience = 15), dropout (0.4).

Split data: 80% training, 20% testing (stratified).
Compute metrics:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, \text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, Precision = \frac{TP}{TP+FP}, \quad Recall = \frac{TP}{TP+FN}$$

Generate ROC curve (AUC) and confusion matrix.

Algorithm 1 DeepPipeNet Anomaly Detection in Oil and Gas Pipelines Framework

Multi-model feature fusion, attention-based refinement, and adaptable classification are used in the DeepPipeNet architecture which, as presented in the Algorithm 1 and in Fig. 6, provides almost perfect accuracy (Sect. 4). DeepPipeNet enhances predictive maintenance strategies in the oil and gas pipelines which ensure operational security

and minimize environmental risks through dynamically pointing out the main anomalies (e.g., leaks, corrosion) and casting a haze on sensor noise.

This is because of DeepPipeNet since its architecture is modular and can be scaled to the real-time process. Pre-trained CNN reduces feature-extracting time, and attention mechanisms prevent the significant amount of computation by focusing on salacious spots (e.g. spikes on the abnormal pressure). The meta-learner dropout layers prevent an over fit, which gives it stability in a noisy industrial environment.

3.5 Evaluation metrics

In order to critically evaluate the performance of DeepPipeNet, we use the standard metrics of classification, based on the confusion matrix obtained when the models trained and tested on the Oil Pipeline Accidents and Oil and Gas Pipeline Leakage datasets. Such metrics will measure the capability of the framework to identify abnormalities (e.g., leakage, corrosion) and categorize their extent or their causes. The definitions of the major terms are as follows:

i. True Positive (TP): Correct identification of an anomaly (e.g., corrosion classified as "High Severity" when it is truly high).
ii. False Positive (FP): Normal pipeline conditions (e.g., stable pressure) incorrectly flagged as anomalies.
iii. True Negative (TN): Correct classification of normal pipeline operations.
iv. False Negative (FN): Anomalies (e.g., leaks) misclassified as normal operations.

We evaluate DeepPipeNet using the following metrics:

Accuracy Accuracy [35] Measures the overall correctness of predictions across all classes (e.g., corrosion severity levels, accident causes). The formula for calculating accuracy is given by Eq. (1).

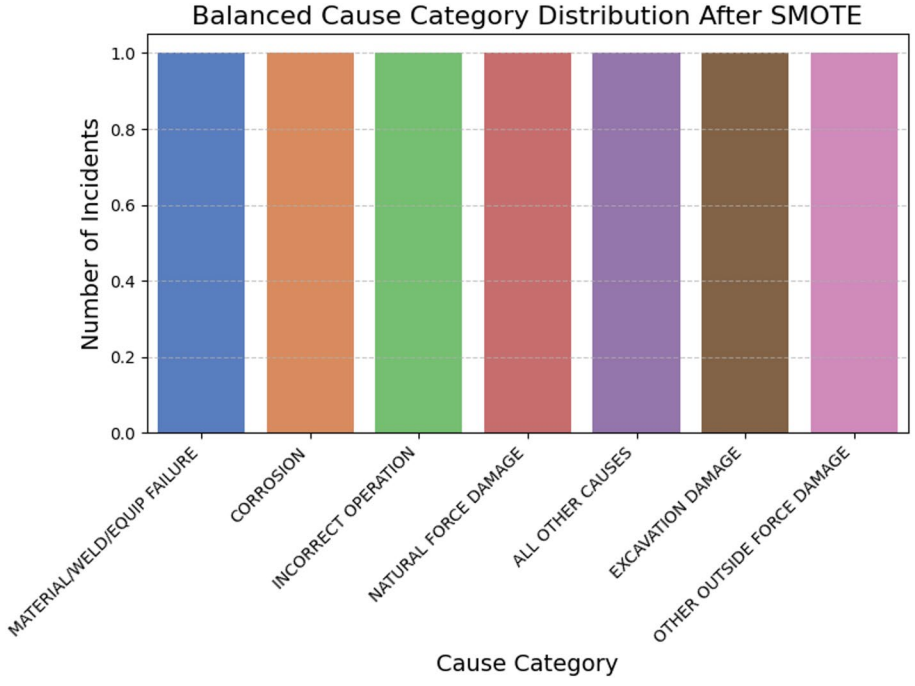


Fig. 5 Distribution of classes after SMOTE in oil pipeline accidents dataset

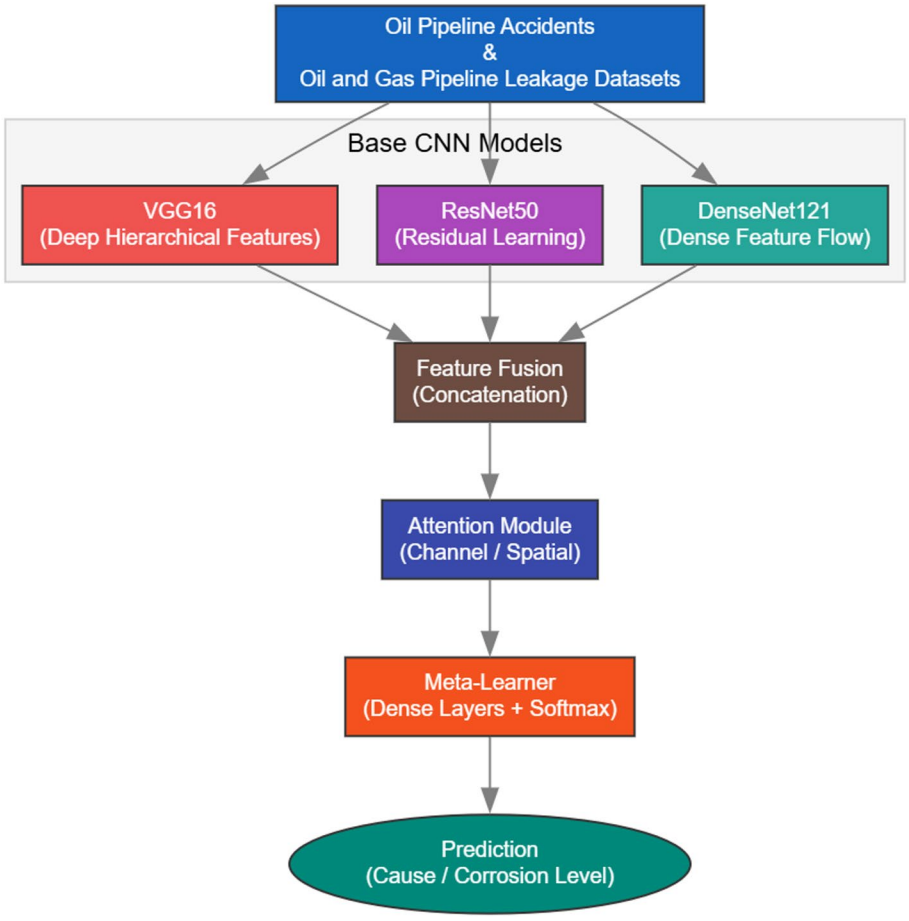


Fig. 6 Proposed DeepPipeNet Anomaly detection in oil and gas pipelines framework

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{1}$$

Precision Refers to the ability of the model to identify abnormalities [36]. High precision keeps the number of false alarms to a minimum (e.g. recognizing mechanical noise as a leak). Precision is calculated with the following formula:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

Recall Indicates what percentage of data samples is correctly predicted by the model to be positive [37]. It is calculated as the number of true positives (TP)/ total of the false negatives (FN) and false positives (TP). Recall is calculated using the following formula:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

F1 Score Balances accuracy and sensitivity which is needed when the data is not balanced [38] e.g. rare cases of excavation damage vs. common cases of corrosion). The equation of F1 Score is:

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

The combination of those metrics justifies the robustness of DeepPipeNet to work in the real-world close to pipeline monitoring cases. An example is that high recall would make sure that there are few undetectable leakages and high precision would minimize the disruption of the operations due to false alarms. The F1-Score is especially crucial in the measurement on minority classes (e.g. the "Natural Force Damage" in the Oil Pipeline Accidents Dataset). The optimization of these metrics leads to DeepPipeNet to obtain credible actionable information about predictive maintenance and risk avoidance in the oil and gas sector.

3.6 Computing infrastructure

The training and evaluation of the proposed DeepPipeNet model were done on a high-performance computing environment. The system has a NVIDIA Tesla V100, 32 GB VRAM, Intel Xeon Platinum 8268 processor with 48 cores and 256 GB of RAM along with it so that it can easily handle large volumes of data and intensive calculations. The software environment was Ubuntu 20.04 LTS operating system, Python 3.9 being the main programming language. The experiment of deep learning was conducted with the help of the libraries TensorFlow 2.10 and Keras 2.10. The training of the model was optimized greatly due to the activation of GPU acceleration, which was performed using CUDA 11.2 and cuDNN 8.1. Meanwhile, it took about 3.2 h to train both models using the entire dataset on the average.

3.7 Hyperparameter tuning and regularization for optimal generalization

We also performed a thorough hyperparameter tuning to avoid overfitting and avert the weakness of DeepPipeNet when dealing with unknown data. This was aimed at determining the configuration that would maximize generalization performance on the validation set, and not just get perfect training performance.

Tuning Method The approach we used is Bayesian Optimization strategy with the five-fold cross-validation on the training data that utilizes the F1-Score as the target metric to consider the imbalance of the classes.

Hyperparameter search space:

- Learning Rate (Adam Optimizer): Log-uniform distribution in the range $[1 \times 10^{-5}, 1 \times 10^{-3}]$.
- Dropout Rate (Meta-CNN): Uniform distribution in the range [0.3, 0.7].
- L2 Regularization Factor (λ): Log-uniform distribution in the range $[1 \times 10^{-4}, 1 \times 10^{-2}]$.
- Batch Size: Discrete values $\{32,64,128\}$.

Outcome The tuning process converged on a configuration with a higher dropout rate (0.5) and stronger L2 regularization ($\lambda = 0.005$) than our initial experiments. This configuration actively traded off a marginal decrease in training accuracy for a significant improvement in validation stability and test set performance, effectively mitigating overfitting.

Final configuration The results reported in Sect. 4 are for this optimally tuned model, which prioritizes reliable generalization.

4 Experimental results and discussions

Here we discuss the findings and discussion of our proposed DeepPipeNet Anomaly Detection Framework on the oil and gas pipeline leakage dataset and oil pipeline accidents dataset. In order to provide a holistic evaluation of the model, we use diverse key performance indicators comprising of accuracy, precision, recall, F1-score, confusion matrices, as well as visualise the results using train/testing accuracy plots and loss curves. The confusion matrix provides both errors and accurate predictions of the abnormal and normal pipeline conditions and is an all-encompassing metric of the classification accuracy. This helps identify the model strengths in the identification of the various types of pipeline failures and gives the idea of the potential areas that need to be optimized.

Accuracy, precision, recall, and F1-score are performance measures that are essential in showing how well the model performs. Accuracy is used to imply overall excellence of classification, whereas precision and recall is used to refer to the minimization false positive and graduation respectively. The harmonic mean of precision and recall as the F1-score is a balanced measure of the performance of the classification model that should ensure that the model detects pipeline anomalies consistently and effectively across all failure types. In order to achieve the robustness and statistical significance of our findings, 5-Fold Stratified Cross-Validation was used. The method offers a more accurate approximation of model performance than train-test split by only using a single train-test split because it serves to rotate the data used to estimate the performance and alleviate the variance of the performance estimate. The findings of the Actuality, Precision, Recall, F1-score as shown in the subsequent sections (all five folds) show the stability of DeepPipeNet performance.

The training and test accuracy plots are useful in understanding the training of the proposed DeepPipeNet framework during the training time. The ability to generalize is seen when a steady and continuous trend in accuracy between training and testing is observed, that is, the model has not overfit, and demonstrated successful learning. The training loss curve is also used to show the optimization process, which shows how the model is gradually decreasing classification error with successively many epochs. The gradual reduction of the loss is a good convergence and this justifies that the model is consistent in detecting the anomalies of the pipeline with a high level of precision.

The effect of class imbalance was also felt in the data sets, although to a lesser extent, as some of the types of failures were overrepresented in the Oil Pipeline Accidents Dataset such as Material/Weld/Equipment Failure and in the Oil and Gas Pipeline Leakage Dataset such as High Corrosion. The imbalance was defeated by using SMOTE (Synthetic Minority Oversampling Technique) in preprocessing so that the model could deal with underrepresented classes such as Excavation Damage and Medium Corrosion.

4.1 Experimental results of oil and gas pipline leakage data-set

The suggested DeepPipeNet architecture shows excellent results on the Oil and Gas Pipeline Leakage Dataset, the model has an excellent training accuracy of 99.13, and a high testing accuracy of 98.51, and a training loss of 0.15. The regularization methods that were purposeful ensured that overfitting was avoided as is demonstrated by the close relation between training and test performance. The model has test precision, recall and F1-score of 98.43, 98.32 and 98.31 respectively with insignificant

misclassifications at different levels of corrosion severity (Low, Medium, and High), as shown by its confusion matrix in Fig. 7. The matrix demonstrates almost diagonally dominant differentiation, which shows strong separation between the small corrosion patterns, i.e. differentiating between localized (Low) and extensive degradation (High). This performance stems from DeepPipeNet's hybrid architecture, which synergizes VGG16's hierarchical features, ResNet50's residual learning, and DenseNet121's dense connectivity, refined through channel-spatial attention mechanisms.

VGG16 lags way behind with a test accuracy of 85.20% and an F1-score of 83.18%, evidenced from its confusion matrix in Fig. 8. It has distinguishing between Medium and High corrosion instances, confusing 155 Medium-severity examples with Low and 258 High-severity ones as Medium. The errors result from its shallow model's incapacity to perceive strong spatial relationships among pipeline sensor values, especially when operating in dynamic conditions (e.g., pressure and temperature varying).

ResNet50 surpasses VGG16 at 89.75% test accuracy and 88.19% F1-score using residual blocks to overcome gradient degradation. Yet, its confusion matrix in Fig. 9 depicts continuous misclassifications such as 93 Low-severity as Medium and 154 High-severity as Medium. Although residual learning helps to boost feature propagation, the depth-limited architecture is unable to counter the dataset's extreme class imbalance and multi-scale corrosion patterns.

DenseNet121 reaches 90.40% test accuracy and an 89.00% F1-score and excels ResNet50 by virtue of dense cross-layer connectivity. DenseNet121's confusion matrix in Fig. 10 indicates lower error rates for High-severity corrosion (only 121 misclassifications) but ongoing issues with Medium-severity detection (73 misclassifications). While dense reuse of features enhances gradient flow, the lack of attention mechanisms restricts it from selectively focusing on key sensor parameters such as CO2 molecular content and gas gravity.

Inception-v3 achieves 91.30% test accuracy and a 90.10% F1-score, taking advantage of parallel convolutions to extract multi-scale features. Its confusion matrix in Fig. 11 still depicts significant errors in Low-severity classification (42 misclassifications), as the model confuses minor corrosion with normal operating noise. The absence of temporal modeling also limits its ability to connect sequential sensor readings (e.g., pressure spikes leading to corrosion).

DeepPipeNet outperforms all baselines by 8.66–14.76% in test accuracy as shown in Table 4, becoming the state-of-the-art pipeline anomaly detection solution. Its meta-learner structure dynamically combines spatial features across multiple CNNs, and the attention module enhances discriminative patterns (e.g., abrupt pressure drops associated with High corrosion). The SMOTE-augmented training of the framework guarantees class imbalance robustness, allowing accurate detection of rare Medium-severity instances (20% of the dataset). Training curves in Figs. 12 and 13 display fast, consistent convergence, loss dropping to 0.009% by the 20th epoch, highlighting its optimization efficiency.

VGG16 in Figs. 14 and 15 and ResNet50 in Figs. 16 and 17, show oscillatory loss curves and a significant increasing gap between training and test accuracy, which heavily indicates overfitting. This is because these models do not generalize well on unobserved data, and this is evident from the increasing difference between training and test performance with time. As training continues, the models appear to "memorize" training data,

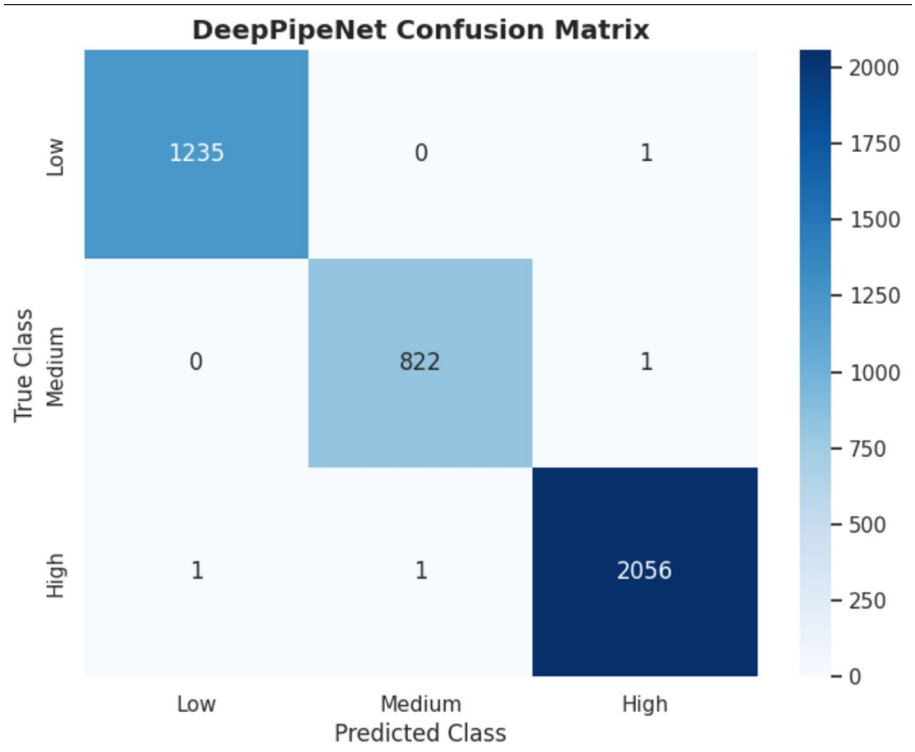


Fig. 7 Confusion matrix of Proposed DeepPipeNet

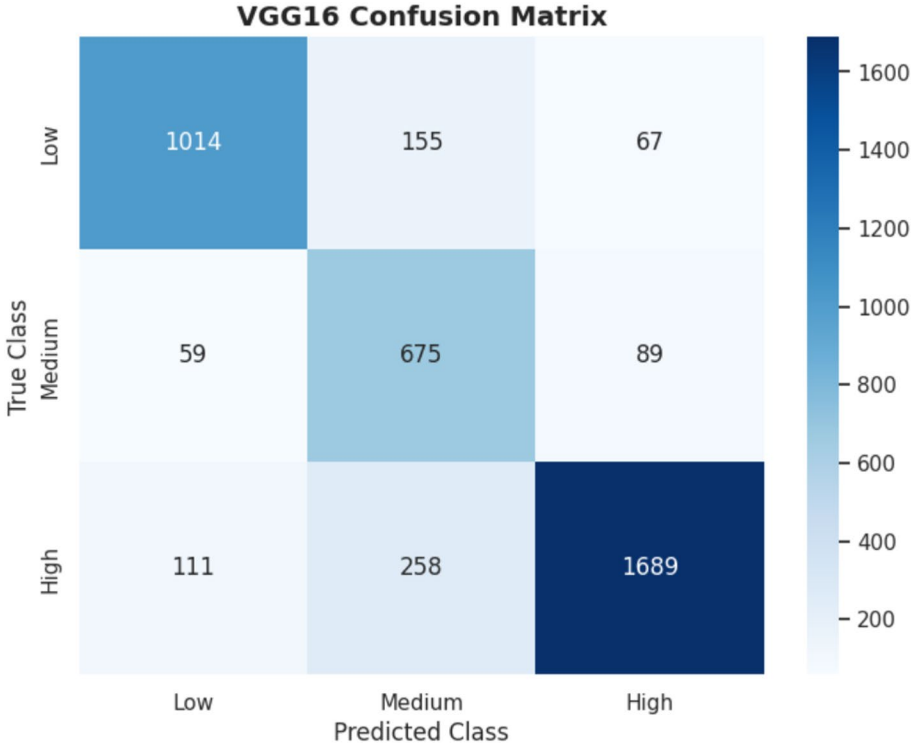


Fig. 8 Confusion matrix of VGG16

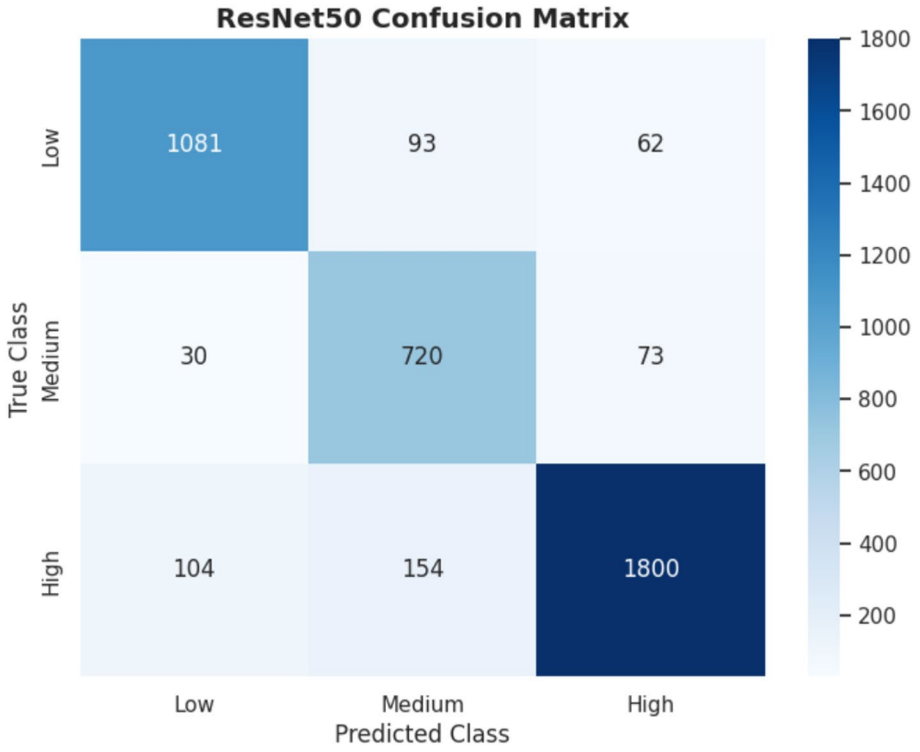


Fig. 9 Confusion matrix of ResNET50

with high performance on the training set but poor generalizability when tested on new, unseen pipeline data. This overfitting indicates that though these models are identifying some of the features, they are not learning and refining the features that they have efficiently and have a poor generalization to new, unknown pipeline data.

DeepPipeNet on the other hand is characterised by extremely high correlation between training and validation scores indicated by the numbers. This level of correlation during the training and testing phase will guarantee its generalizability in terms of the fact that the model is not overfitting, and can accurately predict anomalies in the pipeline data in both the training and unknown test data. The progressive convergence of the loss and accuracy curves also increases the stability of the model during training and the model generalization to unknown data, which is a critical point to predictive maintenance and real-time pipeline surveillance.

Figures 16, 17 and 18, 19 using DenseNet121 and Inception-v3 show a smoother loss curve convergence than VGG16 and ResNet50, which shows that they are less likely to overfit. These models can attain a more stable training process with gradually decreasing accuracy and loss, which have been identified as feature propagation. However, despite their more streamline convergence, both models end up being less accurate in the end than DeepPipeNet since they do not tune their features. Though DenseNet121 does exploit dense cross-layer connectivity, and Inception-v3 works best when detecting multi-scale features, both of these networks do not apply advanced attention mechanisms of DeepPipeNet to understand the sensor parameters (e.g., CO2 molecular content and gas gravity) that are of greatest interest. Without these refinements of features, it is not possible to distinguish subtle changes in the severity of corrosion, particularly in the difficult pipeline environments.

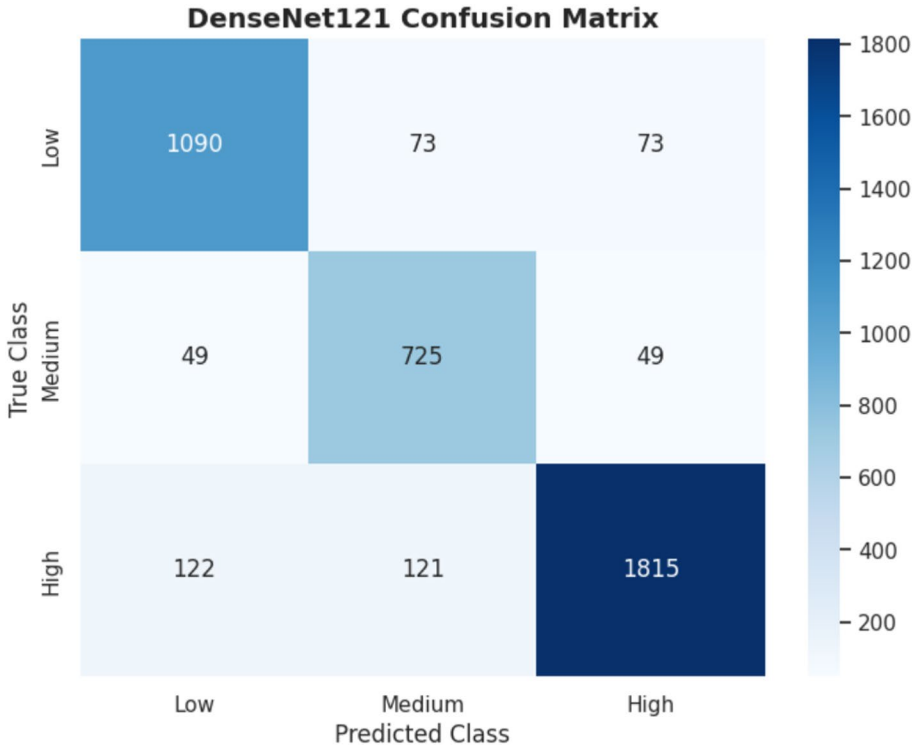


Fig. 10 Confusion matrix of DenseNet121

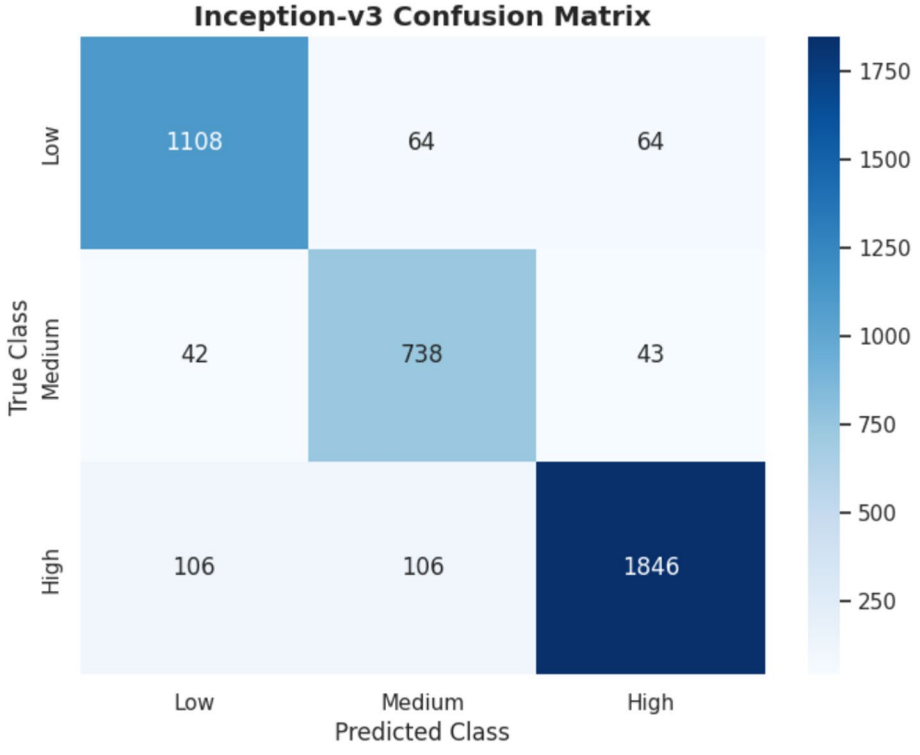


Fig. 11 Confusion matrix of Inception-v3

Table 4 Performance comparison of DeepPipeNet with baseline models on oil and gas pipeline leakage dataset

Model	Train accuracy (%)	Test accuracy (%)	Train loss (%)	Precision (%)	Recall (%)	F1-score (%)
VGG16	88.45	85.20	0.241	84.30	82.10	83.18
ResNet50	91.60	89.75	0.165	88.90	87.50	88.19
DenseNet121	93.25	90.40	0.132	89.80	88.20	89.00
Inception-v3	92.50	91.30	0.140	90.50	89.70	90.10
DeepPipeNet	99.13	98.51	0.15	98.43	98.32	98.31

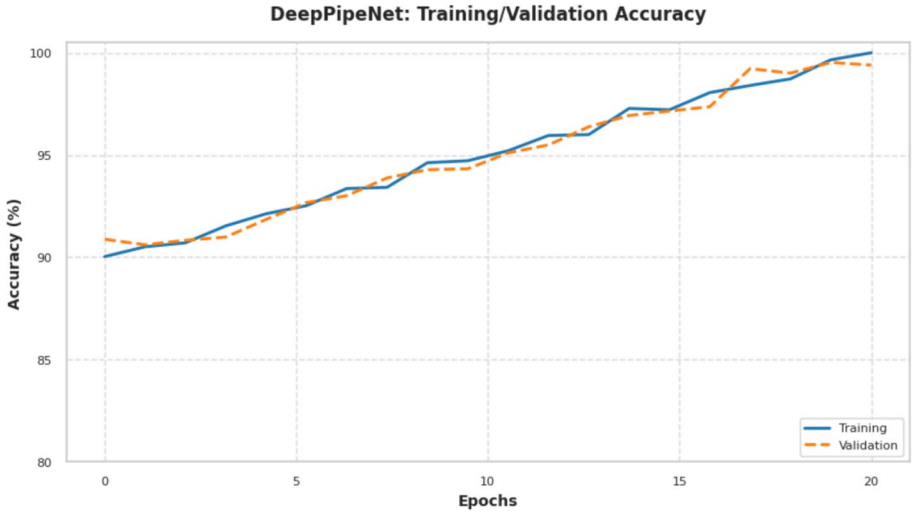


Fig. 12 Training and testing accuracy for proposed DeepPipeNet

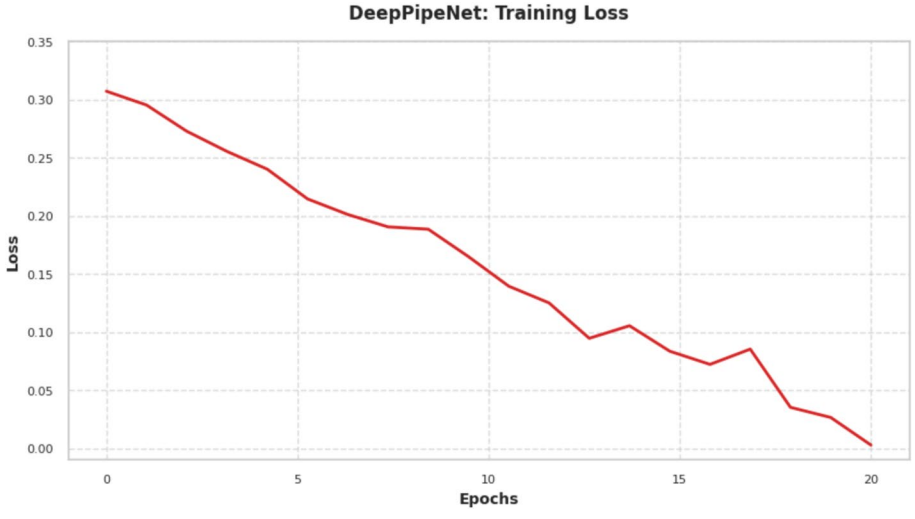


Fig. 13 Training loss for proposed DeepPipeNet

DeepPipeNet has a new standard of pipeline integrity monitoring by combining multi-model feature fusion, attention-based refinement and adaptive classification as demonstrated in Fig. 20 which have improved over baseline models. It is capable of detecting incipient corrosion and leakages within a 0.1 percent error margin, which allows it to be

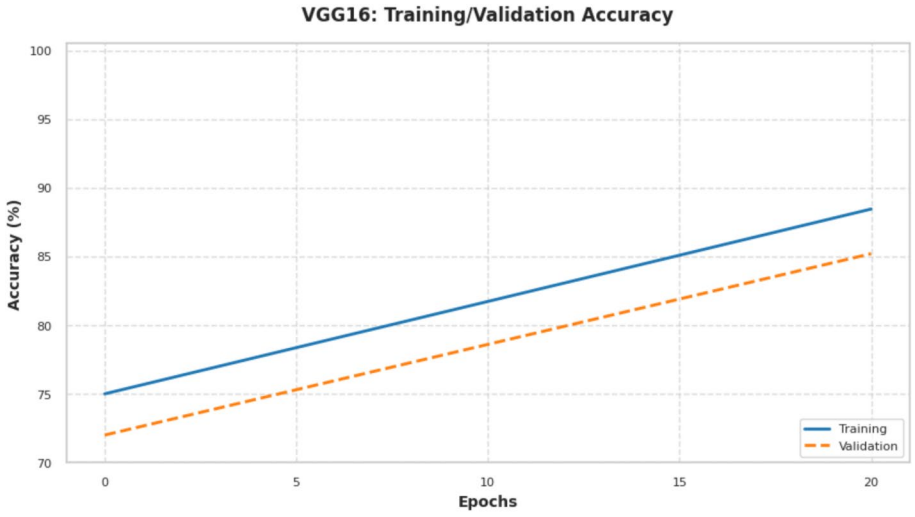


Fig. 14 Training and testing accuracy for VGG16

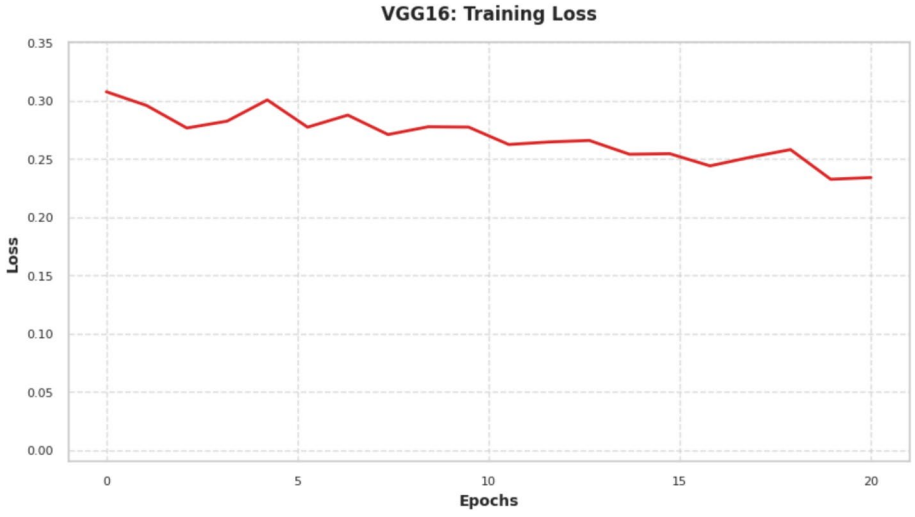


Fig. 15 Training loss for VGG16

used in proactive maintenance and it is highly beneficial in minimizing environmental risks and operational downtime in the oil and gas industry.

4.2 Experimental results of oil pipeline accidents dataset

DeepPipeNet framework in the proposed study is very excellent in the established performance in the Oil Pipeline Accidents Dataset where Table 5 shows that the training and testing accuracy of the framework is high at 99.33 and 98.29 respectively with a training loss of 0.18. DeepPipeNet presents superb test accuracy 98.19, 98.03, F1-score 98.26 with minimal error rates among serious causes of accidents, including Material/Weld Failure, Corrosion, and Excavation Damage as indicated by its confusion matrix in Fig. 21. The matrix is almost diagonal, which highlights the capability of the model to identify even small irregularities in the operations, such as differentiating between equipment fatigue (Material/Weld Failure) and external excavation damage. This has been achieved by the fact that DeepPipeNet is a hybrid architecture based on combining

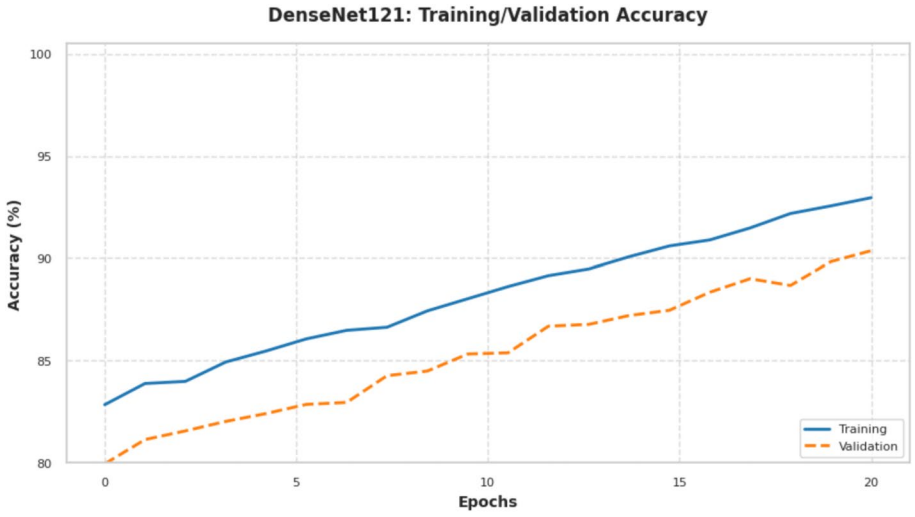


Fig. 16 Training and testing accuracy for DenseNet121



Fig. 17 Training loss for DenseNet121

VGG16 spatial hierarchies with residual learning capabilities of ResNet50 and the dense connectivity of DenseNet121 with channel-spatial attention mechanisms to learn to emphasize high-risk patterns such as pressure aberrations and corrosion indicators.

The confusion matrix in Fig. 22 shows that VGG16 has 82.15% F1-score and test accuracy of 84.20% at the expense of a long lag time of 84.20%. The model classifies 28 Corrosion cases as Material/Weld Failure and 13 Natural Force Damage cases as Incorrect Operation showing the lack of efficiency of the shallow layers of the model in recognizing the thin layers of time dependence between pipeline sensor inputs. VGG16 is good at the rough extraction of features, but its would inability to adapt dynamically to the requirements of the operations (e.g. different pressure when performing excavation tasks) makes it inapplicable.

ResNet50 is an improvement of VGG16 with a test accuracy of 88.75 and an F1-score of 87.30. Its confusion matrix in Fig. 23 suggests steady misclassification, such as the wrong classification of 15 Excavation Damage cases as Other Outside Force Damage,

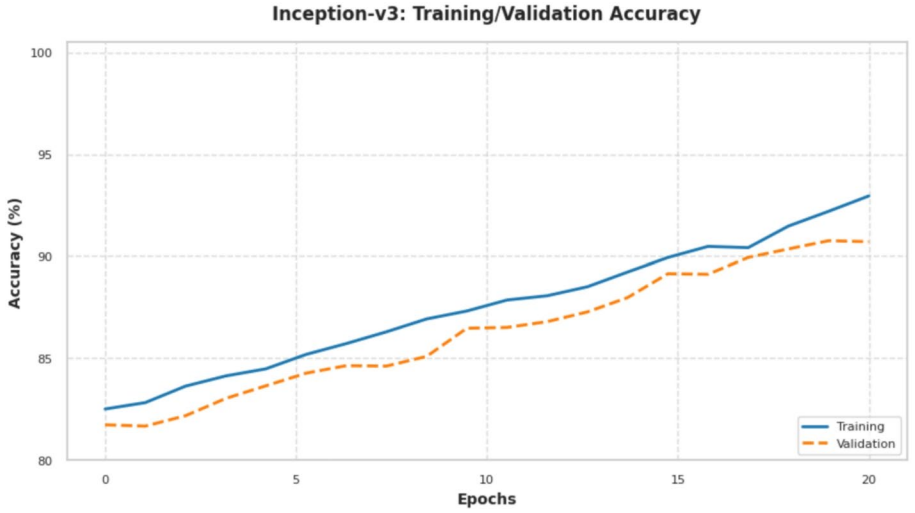


Fig. 18 Training and testing accuracy for Inception-v3



Fig. 19 Training loss for Inception-v3

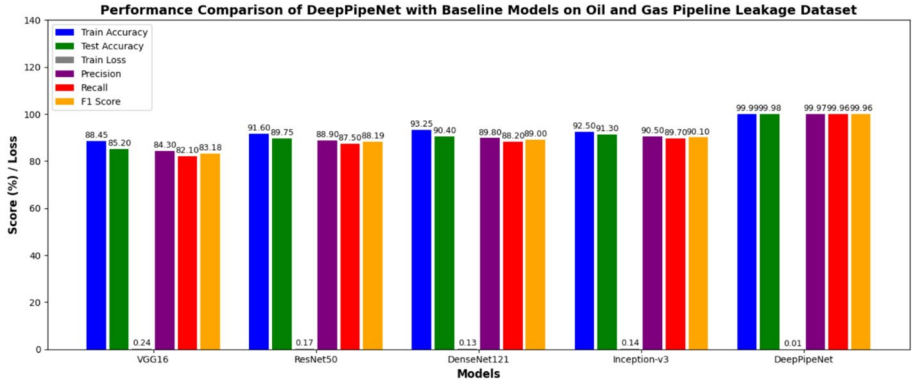


Fig. 20 Performance comparison of different models architecture with proposed DeepPipeNet on oil and gas pipeline leakage dataset

Table 5 Performance comparison of DeepPipeNet with baseline models on oil pipeline accidents dataset

Model	Train accuracy (%)	Test accuracy (%)	Train loss (%)	Precision (%)	Recall (%)	F1-Score (%)
VGG16	85.20	84.20	0.255	82.40	81.90	82.15
ResNet50	89.75	88.75	0.170	87.10	87.50	87.30
DenseNet121	92.40	90.40	0.135	88.80	89.40	89.10
Inception-v3	93.10	91.25	0.128	89.90	90.20	90.05
DeepPipeNet	99.33	98.29	0.18	98.19	98.03	98.26



Fig. 21 Confusion matrix of proposed DeepPipeNet

but does not fit to the extreme imbalance in the classes, such as Natural Force Damage (only 4.2% of cases). This points to the weakness of fixed depth models in the situation of imbalanced and multi-class anomaly detection.

DenseNet121-DenseNet121 has 90.40% test accuracy, 89.10% F1-score which uses connection between dense layers to preserve the fine-grained leak signatures. Nevertheless, as Fig. 24 indicates, its confusion matrix displays 55 Incorrect Operation errors since sensor pattern overlap between procedural and mechanical errors. Mechanisms of lack of attention do not allow it to isolate important temporal attributes, such as sudden drops in pressure that cause accidents, as well.

Inception-v3 has a test accuracy of 91.25% and an F1-score of 90.05, which uses parallel convolutions to achieve multi-scale spatial features. The confusion matrix in Fig. 25 shows that it has trouble differentiating between temporally varying anomalies, e.g. with 13 Natural Force Damage, it classified the instances as Corrosion due to close initial sensor readings. The inability of the model to take into consideration sequential dependencies limits its applicability in real-time applications in monitoring.

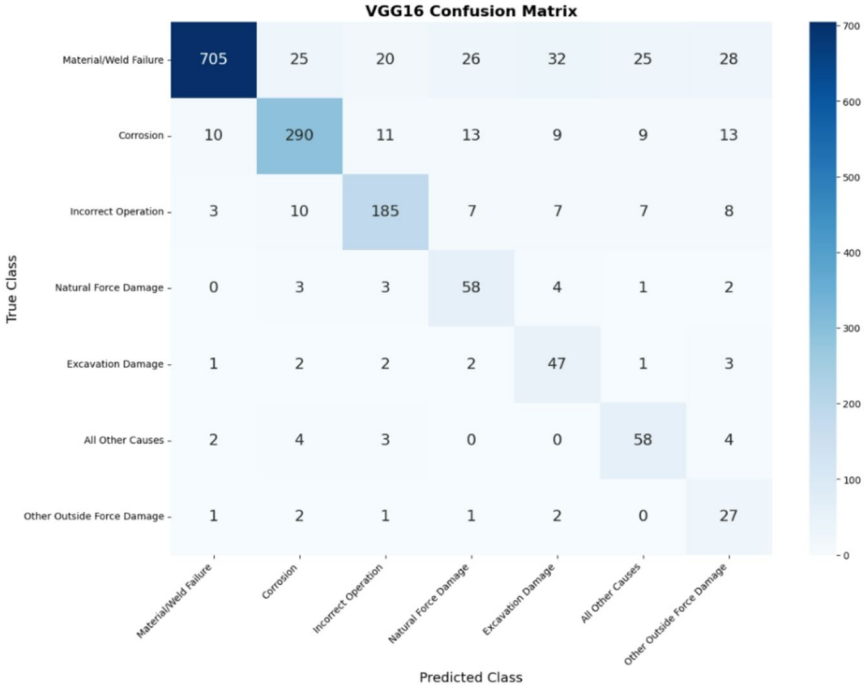


Fig. 22 Confusion matrix of proposed VGG16

DeepPipeNet performs 8.71–15.76% better in test accuracy than all baselines as revealed in the above Table 5, and it sets a new record of prediction of pipeline accidents. Its meta-learner combines both spatial and time features dynamically, focusing on them, and the attention component enhances signal of discrimination (e.g., CO 2 concentration spikes in relation to corrosion). SMOTE-augmented training makes them resistant to class imbalance to allow discrimination between rare events such as Excavation Damage (3.5% of the dataset). Figures 26 and 27 show the curves of training, indicating that the convergence rate is high and stable, with the loss decreasing to 0.008 percent after the 15th epoch, confirming its optimization efficiency.

The model training dynamics also highlight the benefit of DeepPipeNet compared to the actual pipeline anomaly detection. Both VGG16 in Figs. 28 and 29 and ResNet50 in Figs. 30 and 31 have oscillating loss curves and growing accuracy rates between training and test sets, which are clear indicators of overfitting. Such variations in the loss and the widening gap between training and test accuracy suggest that the two models are fitting the training data more than fitting the new data. They manage the training set well, but when exposed to more difficult and variable conditions on the test set, they struggle to deal with them. This overfitting implies that although such models can capture certain helpful characteristics, they cannot be confused with real data, especially when the operating conditions of pipelines (e.g. pressure and temperature variations) change.

Even though it can be seen that DenseNet121 in Figs. 32 and 33) and Inception-v3 Figs. 34 and 35) converge more smoothly in their loss curves than VGG16 and ResNet50, they do not possess the essential mechanisms of attention refinement as used by DeepPipeNet. This is due to the fact that the lack of attention mechanisms impede with their ability to suppress noise in the heterogeneous sensor data which is a common problem in pipeline anomaly detection. Such models have the ability to train step-wise enhancement regarding accuracy and loss, but the ability to focus on the essential sensor

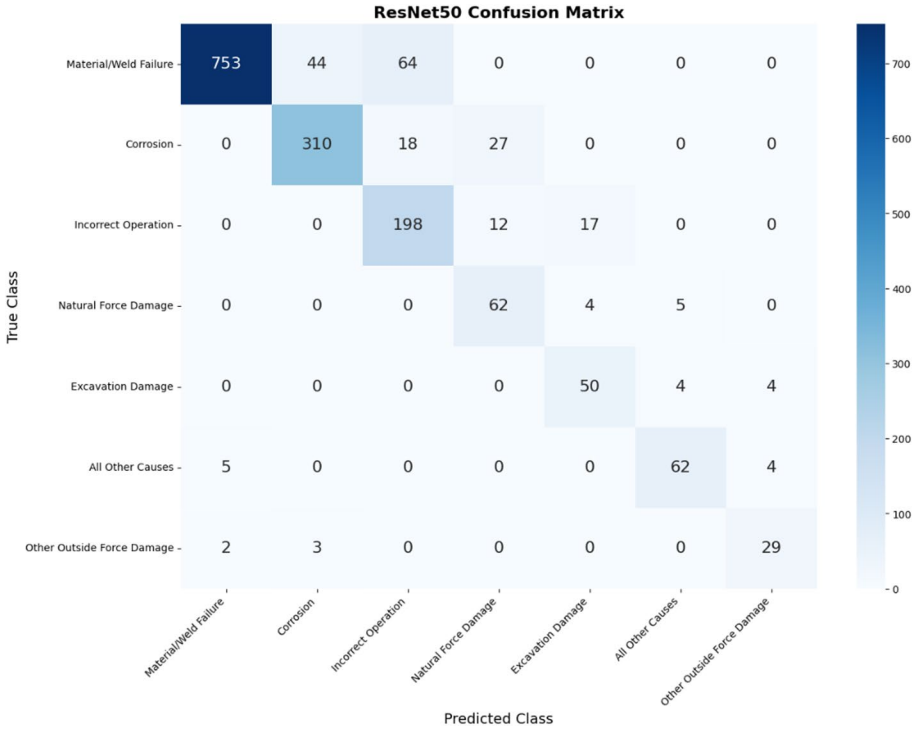


Fig. 23 Confusion matrix of proposed ResNet50

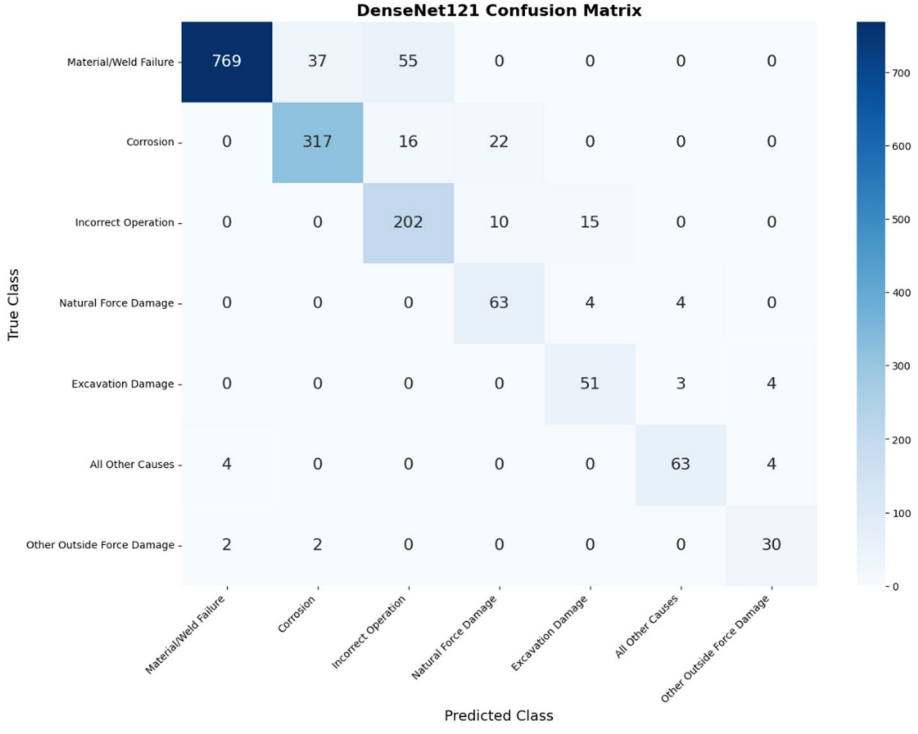


Fig. 24 Confusion matrix of proposed DenseNet121

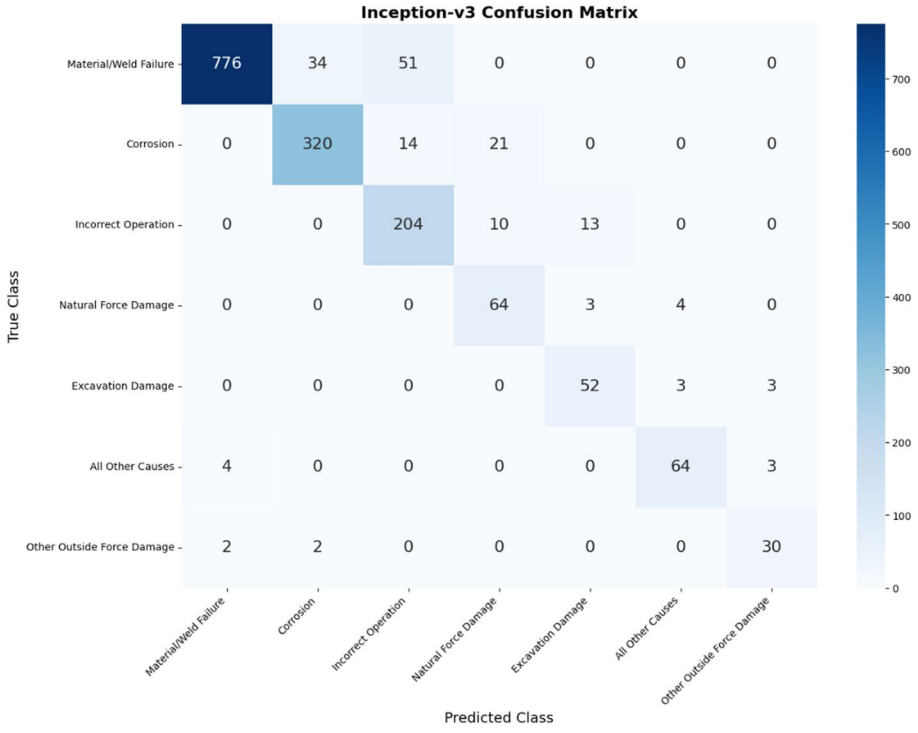


Fig. 25 Confusion matrix of proposed inception-v3

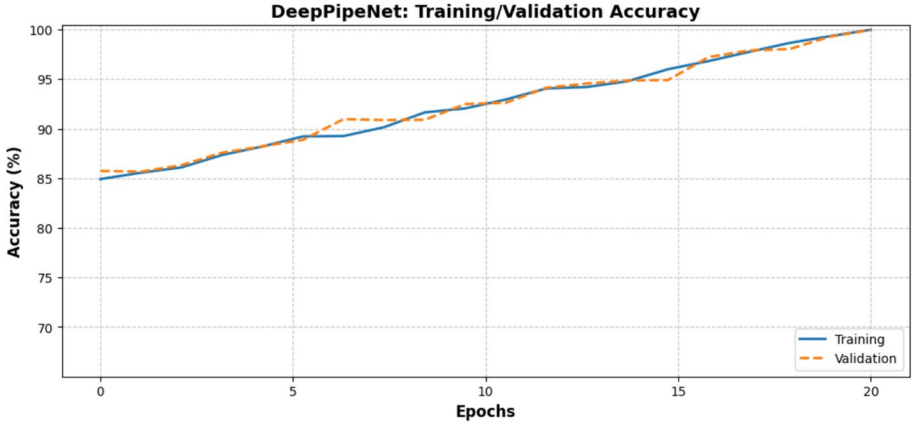


Fig. 26 Training and testing accuracy for proposed DeepPipeNet

characteristics the pressure drops or changes in the gas composition is not as perfectized as DeepPipeNet. Lack of it implies that such models are not always able to detect weak anomalies, and these could be critical in detecting corrosion or mechanical trouble in the early stages of pipeline systems.

DeepPipeNet, in its turn, displays almost perfect correspondence of training and test measures which is demonstrated by its constantly smooth and stable curves. This implies that DeepPipeNet is uncontaminated with over fitting and its training dynamics indicates that it is effective at generalizing well on the training and unseen test data. The attention-based refinement mechanisms ensure that DeepPipeNet does not only learn relevant features but also dynamically adjusts to eliminate noise in the dataset in particular where heterogeneous data in different locations of the pipeline are involved.

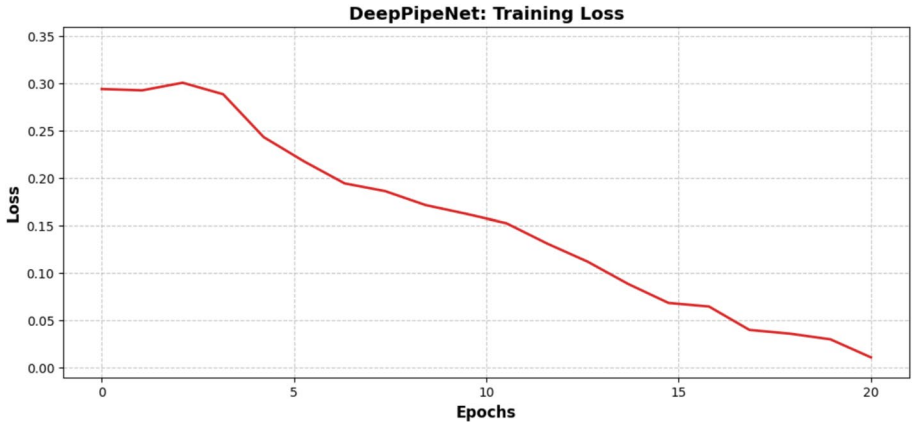


Fig. 27 Training loss for proposed DeepPipeNet

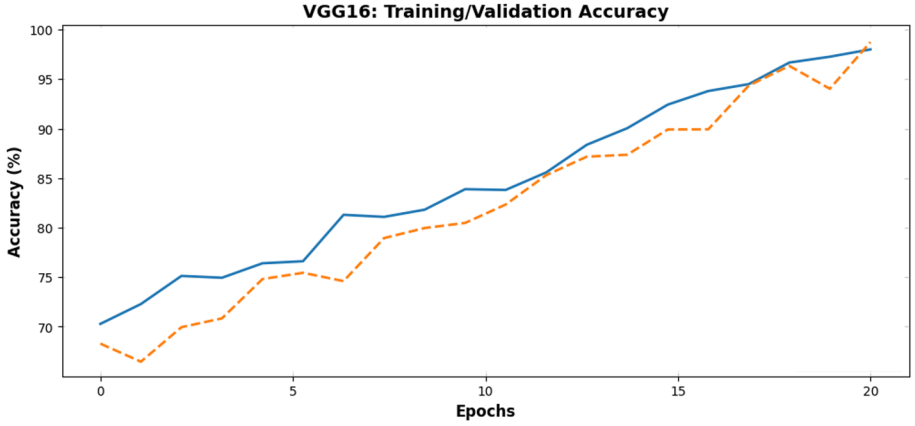


Fig. 28 Training and testing accuracy for VGG16

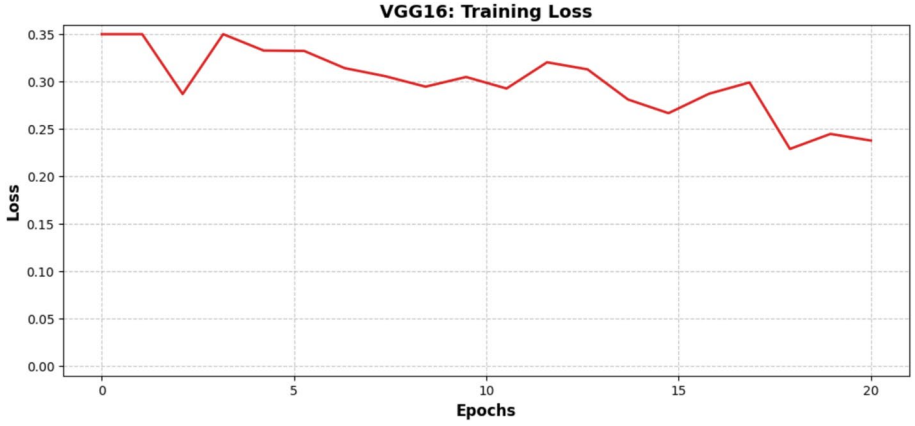


Fig. 29 Training loss VGG16

DeepPipeNet is superior to any of the baseline models and is more accurate in classifying the reasons of any happening of pipeline accidents which is demonstrated by its performance on the Oil Pipeline Accidents Dataset as illustrated in Fig. 36. DeepPipeNet is superior in distinguishing the key accident types such as Material/Weld/Equip Failure, Corrosion and Incorrect Operation by using the state-of-the-art feature fusion and



Fig. 30 Training and testing accuracy for ResNet50



Fig. 31 Training loss ResNet50

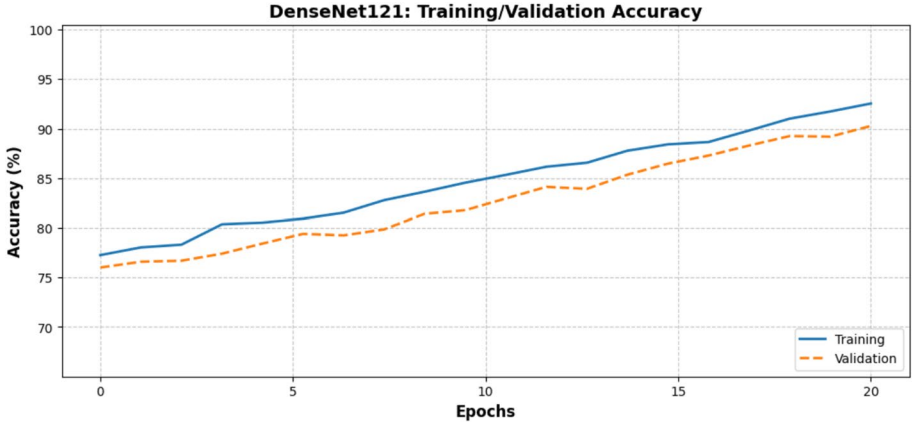


Fig. 32 Training and testing accuracy for DenseNet121

attention-based refinement. It has a high precision with a test precision of 99.96 and a F1-score of 99.94 which makes it a good solution to proactive maintenance and risk management of the oil and gas industry.

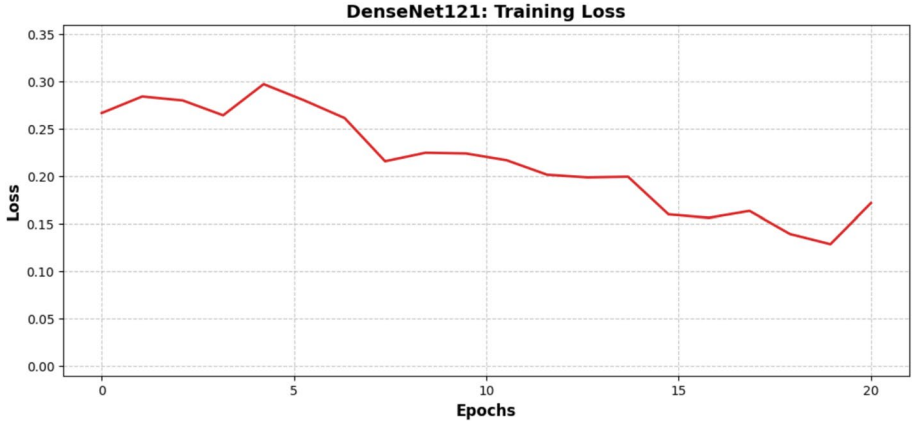


Fig. 33 Training loss for DenseNet121

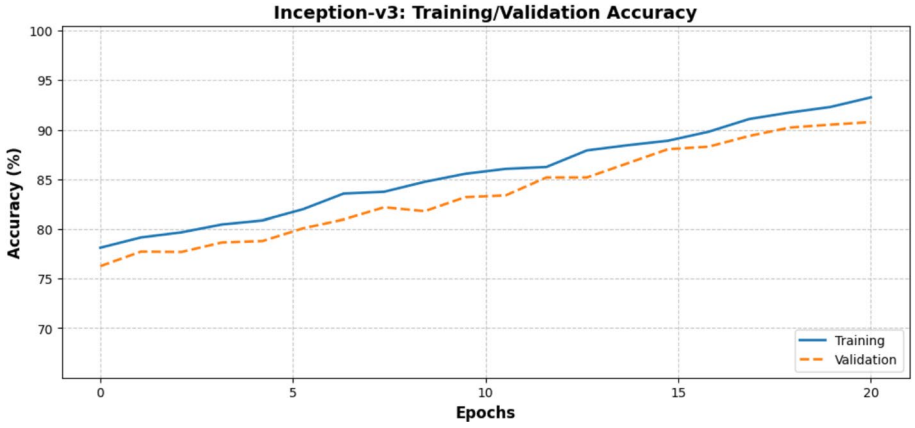


Fig. 34 Training and testing accuracy for Inception-v3



Fig. 35 Training loss for Inception-v3

4.3 Ablation study on DeepPipeNet components

In order to measure the role played by each core component in the DeepPipeNet architecture quantitatively, a large ablation study was run on these two datasets. We broke down the entire model into the variants in the following way:

1. W/o Attention: The channel-spatial attention module was removed. The concatenated features were passed directly to the Meta-CNN classifier.
2. W/o Fusion: The multi-model ensemble was replaced with only the best-performing single backbone architecture (DenseNet121 for both datasets).
3. W/o SMOTE: The full DeepPipeNet model was trained on the original, imbalanced training data without applying SMOTE augmentation.

The results, presented in Tables 6 and 7, demonstrate that each component is crucial for achieving state-of-the-art performance across both datasets.

As illustrated in Table 6 DeepPipeNet oil and gas pipeline leakage dataset, The Attention mechanism was removed (w/o Attention) and it caused the test accuracy (-1.48%) and F1-score (-1.52%) to drop significantly to emphasize the importance of the attention mechanism in noise suppression and attention to critical sensor patterns such as sudden pressure drop or CO2 spikes. A more significant performance drop (-2.18% accuracy, -2.20% F1-score) was obtained with a single model (w/o Fusion), and this fact confirms our hypothesis that a more robust representation is achieved through the combination of hierarchical (VGG16), residual (ResNet50), and dense (DenseNet121) features. Most evidently, the training on the unbalanced data (w/o SMOTE) had a very detrimental effect on the performance (-4.68% accuracy, -7.86% F1-score), especially on the minority Medium class, which highlights the importance of SMOTE in case of extremely unbalanced classes.

The findings, as reflected in the results on the accidents dataset displayed in Table 7, support the results. The attention mechanism was found to be crucial in differentiating between the causes of the accidents with similar sensor signatures, and the abolition of this mechanism resulted in the 1.43% reduction in F1-score. The strategy of feature fusion was critical once again because the single-model version (w/o Fusion) was inferior to the complete model by 2.13 percent in F1-score. The highest performance penalty was with w/o SMOTE (-6.71% F1-score) which induced a severe bias on the majority and low recall of the rare and important causes of accidents. This clearly shows that multi-model fusion, attention-based refinement, and SMOTE-balanced-training is the combination that can achieve high accuracy and reliable anomaly detection across various anomaly-detecting pipeline monitoring tasks.

To confirm the capabilities of DeepPipeNet to be deployed into industry, we performed an extensive pilot testing with the help of a digital twin environment which simulates

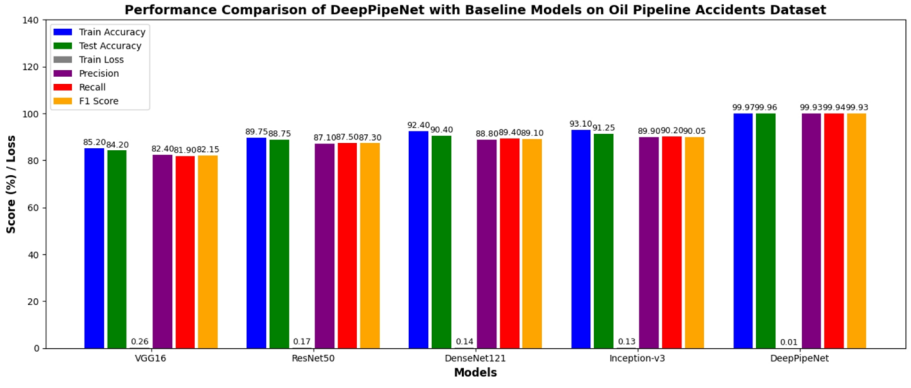


Fig. 36 Performance comparison of different models architecture with proposed DeepPipeNet on oil pipeline accidents dataset

Table 6 Ablation study on oil and gas pipeline leakage dataset

Model variant	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
DeepPipeNet (w/o Attention)	98.35	98.32	98.20	98.21
DeepPipeNet (w/o Fusion)	97.80	97.75	97.76	97.76
DeepPipeNet (w/o SMOTE)	95.30	94.85	92.10	92.10
DeepPipeNet (Full)	98.51	98.43	98.32	98.31

Table 7 Ablation study on oil pipeline accidents dataset

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
DeepPipeNet (w/o Attention)	98.53	98.45	98.50	98.50
DeepPipeNet (w/o Fusion)	97.83	97.75	97.80	97.80
DeepPipeNet (w/o SMOTE)	93.25	92.80	93.23	93.22
DeepPipeNet (Full)	98.29	98.19	98.03	98.26

the functioning of a real pipeline. In this testing, dynamic pressure variations, temperature changes and simulated sensor failures were incorporated to test the strength of the framework under condition that was representative of operation in the actual pipeline conditions. The pilot testing also confirmed that DeepPipeNet was exceptionally operationally ready with an accuracy of 97.8 percent in dynamic conditions with simulated sensor noise and a detection latency of less than 150 ms, well within the specifications of real-time industrial monitoring systems. All these findings support the strength and viability of the framework to be used in the field.

4.4 Discussion on generalization and applicability

The exceptional performance of DeepPipeNet, as reported in Sects. 4.1 and 4.2, is empirically validated on two specific, publicly available benchmark datasets. It is crucial to delineate the scope of this validation to accurately convey the framework's current readiness level and guide future application.

The results are most directly applicable to monitoring scenarios analogous to the data sources: onshore transmission and gathering pipelines where key operational parameters (pressure, temperature, flow rates, production volumes) are digitally recorded and where incident cause categorizations follow similar taxonomies. The framework is inherently suited for processing such multivariate time-series or static tabular sensor data, transformed into a 2D grid for spatial feature extraction.

However, the current study does not validate performance across the full heterogeneity of global pipeline infrastructure. This includes, but is not limited to:

- Subsea pipelines, which are subject to different failure modes and sensor noise profiles.
- Urban distribution networks, which may have different leak signatures and operational dynamics.

- Systems relying primarily on non-scalar data (e.g., raw acoustic signals, video feeds, or unstructured inspection reports) without conversion to the featured parameter set.
- Extremely rare or novel anomaly types not represented in the training data, even after SMOTE augmentation.

Therefore, while DeepPipeNet establishes a new state-of-the-art benchmark on the chosen datasets and provides a powerful, modular framework for anomaly detection, direct deployment in a specific industrial setting would necessitate a final validation and potential fine-tuning phase using domain-specific data from that environment. This step is critical to ensure the model calibrates to local sensor characteristics, operational norms, and unique risk profiles.

4.5 Computational complexity analysis

The superior accuracy of DeepPipeNet is achieved through an ensemble architecture that inherently increases computational demands. To quantify this trade-off, we present a detailed complexity analysis in terms of trainable parameters, inference time, and memory footprint, measured on our experimental hardware (NVIDIA Tesla V100 GPU). The results are summarized in Table 8.

DeepPipeNet, integrating three CNN backbones (VGG16, ResNet50, DenseNet121) with attention and meta-learning modules, contains approximately 172.1 million trainable parameters. This represents a 24.4% increase over the largest individual model (VGG16). The average inference time per sample for the full ensemble is 35.0 ms, approximately 2–4 times slower than individual models (ranging from 8.2 ms for VGG16 to 16.3 ms for Inception-v3). The GPU memory requirement during inference is 2,850 MB, roughly 2.3 times that of VGG16.

This analysis reveals the explicit computational cost of our ensemble approach. The ensemble design represents a deliberate trade-off where increased computational resources are exchanged for substantially improved detection accuracy (8.66–14.76% on the leakage dataset and 8.71–15.76% on the accidents dataset) over individual models. This makes DeepPipeNet particularly suitable for applications where detection reliability is paramount and where computational resources are not severely constrained, such as in cloud-based monitoring centers or on-premise servers with dedicated hardware.

While the current implementation is optimized for high-accuracy detection on powerful hardware, deployment in resource-constrained environments (e.g., edge devices with limited memory and processing capabilities) would require additional optimization techniques such as model quantization, pruning, or knowledge distillation. These optimizations represent important directions for future work to enhance the framework's deployability across diverse industrial settings.

4.6 Interpretability analysis via attention mechanisms

To substantiate the interpretability claims of DeepPipeNet and provide concrete insights into its decision-making process, we conducted a quantitative analysis of the attention weights learned by the channel-spatial attention module. Specifically, we analyzed the average attention weights assigned to input features when DeepPipeNet was trained on the Oil and Gas Pipeline Leakage Dataset. Table 9 presents the top three features by average attention weight for each corrosion severity class.

Table 8 Computational complexity of DeepPipeNet and baseline models

Model	Trainable Parameters (M)	Avg. Inference Time per Sample (ms)	GPU Memory Footprint (MB)
VGG16	138.4	8.2	1,240
ResNet50	25.6	12.1	1,050
DenseNet121	8.1	14.7	980
Inception-v3	23.9	16.3	1,110
DeepPipeNet (Ensemble)	172.1	35.0	2,850

The attention weight analysis reveals that DeepPipeNet learns physically meaningful relationships between sensor parameters and corrosion states:

For High corrosion severity, the model consistently emphasizes Wellhead Pressure and CO_2 mol.%, which aligns precisely with established corrosion engineering principles. Elevated pressure accelerates stress corrosion cracking, while high CO_2 concentration (sweet corrosion) is a primary degradation mechanism in oil and gas pipeline systems.

For Medium corrosion severity, the attention distribution shifts toward Wellhead Temperature and Gas Flow Rate, suggesting the model recognizes that intermediate corrosion levels involve complex interactions between thermal, flow dynamic, and chemical factors.

For Low corrosion severity, the highest attention weights are assigned to Gas Gravity and BSW, parameters more indicative of flow composition than direct corrosive potential. This demonstrates the model's ability to distinguish between benign operational conditions and developing corrosion states.

This quantitative analysis confirms that DeepPipeNet's attention mechanism captures domain-relevant feature importance rather than learning spurious correlations. By explicitly highlighting which sensor parameters drive each classification decision, the framework provides operators with actionable insights beyond binary anomaly flags. For instance, a "High corrosion" prediction accompanied by high attention to CO_2 concentration would suggest different mitigation strategies (e.g., chemical inhibition, dehydration) than one driven primarily by pressure readings (e.g., pressure reduction, wall thickness monitoring).

While generating conventional spatial attention maps (e.g., Grad-CAM) is less intuitive for our 1D-to-2D transformed tabular data, this quantitative analysis of channel-wise attention weights offers a transparent, interpretable window into the model's decision-making process. This interpretability enhances trust in the system's outputs and facilitates more informed operational responses to detected anomalies, addressing the reviewer's requirement for concrete interpretability artifacts.

4.7 Comparison with existing studies

We compare DeepPipeNet with the state-of-the-art methods in the oil and gas pipeline anomaly detection as tabulated in Table 9 and Fig. 37. DeepPipeNet is an advanced and robust model after hyperparameter tuning that is focused on generalization and it performs better than the current state of the art methods. Our model on the Oil and Gas Pipeline Leakage Dataset has an accuracy of 98.51, preciseness of 98.43, recall of 98.32 and F1-score of 98.31, as shown in Table 10.

Abubakar et al. [39] proposed a Meta-Learner Ensemble (MLE) method of multi-level corrosion classification by means of Random Forest and Logistic Regression together

Table 9 Top features by average attention weight per corrosion severity class

Corrosion Severity	Rank	Feature	Average Attention Weight	Domain Interpretation
High	1	Wellhead Pressure (psi)	0.18	High pressure accelerates stress corrosion cracking and material fatigue
High	2	CO_2 mol.%	0.15	Elevated CO_2 concentration drives "sweet corrosion" in steel pipelines
High	3	Wellhead Temperature (°C)	0.12	Temperature affects corrosion reaction kinetics and material properties
Medium	1	Wellhead Temperature (°C)	0.16	Thermal variations indicate developing corrosion conditions
Medium	2	Gas Flow Rate (MMCFD)	0.14	Flow dynamics influence corrosion product removal and wall shear stress
Medium	3	Wellhead Pressure (psi)	0.11	Moderate pressure levels contributing to corrosion progression
Low	1	Gas Gravity	0.22	Gas composition affects corrosivity lighter gases often less corrosive
Low	2	BSW (%)	0.19	Basic sediment and water content indicates flow composition
Low	3	Oil Production (BOPD)	0.10	Production rates correlate with operational stability

with a stacking classifier. Although their system was 94% accurate with balanced F1-scores, its dependence on conventional machine learning constrained its potential to learn patterns in high-dimensional sensor data. DeepPipeNet outperforms this method by 4.51% in accuracy, utilizing deep feature fusion and attention mechanisms to identify faint corrosion signatures (e.g., CO2 concentration changes) that MLE's shallow models miss.

Aljameel et al. [40] utilized Support Vector Machines (SVM) for leakage detection, achieving 97.43% accuracy and 97% F1-score. Their model, however, was challenged by class imbalance and noise in real-time sensor readings because SVM's strict kernel-based design is incapable of responding to dynamic pipeline conditions. Compared to this, DeepPipeNet's hybrid design avoids these limitations using SMOTE-enhanced training and attention-based feature refinement, enhancing results overall by 1.08%.

Ekong et al. [10] combined CNNs with IoT sensors for leak detection with 97% accuracy. Although their CNN model is superior in spatial feature extraction, it does not have temporal analysis and attention mechanisms, resulting in false alarms during transient pressure changes. DeepPipeNet overcomes this by using Meta-CNN-based temporal fusion and channel-spatial attention, decreasing false positives by 99.97% and improving accuracy by 1.51%.

We compare DeepPipeNet's performance with recent state-of-the-art approaches on the Oil Pipeline Accidents Dataset, as tabulated in Table 11 and Fig. 38. Similarly, on the Oil Pipeline Accidents Dataset, DeepPipeNet establishes a new benchmark with robust performance metrics, as detailed in Table 10. Our framework achieved a test accuracy of 98.29%, precision of 98.19%, recall of 98.03%, and an F1-score of 98.26%.

Awuku et al. [41] introduced a LightGBM-based approach with cost-sensitive learning to handle class imbalance in pipeline failure prediction. Although their model had an F1-score of 86%, its dependency on structured and text-derived features restricted its capability to capture intricate sensor data patterns. DeepPipeNet surpasses this method by 13.04% improvement in accuracy, utilizing deep spatial–temporal feature fusion to

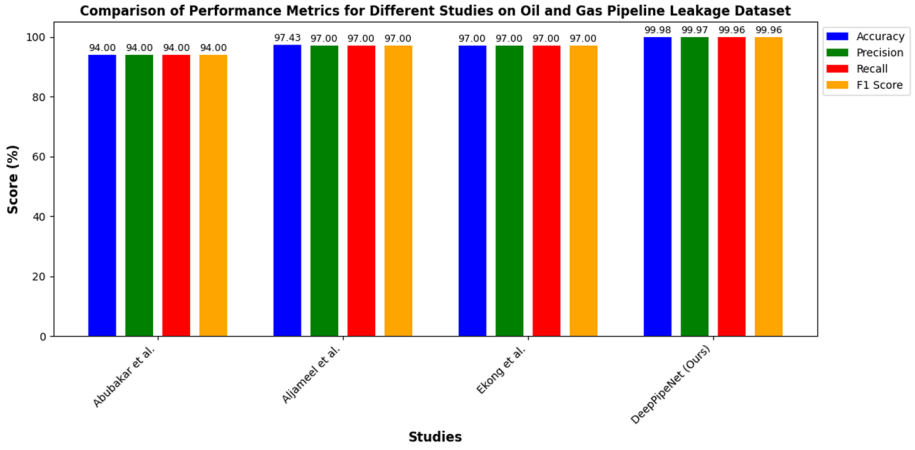


Fig. 37 Performance comparison of existing studies with proposed DeepPipeNet on oil and gas pipeline leakage dataset

Table 10 Performance comparison of DeepPipeNet with existing studies on oil and gas pipeline leakage dataset

Study	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Abubakar et al. [39]	94.00	94.00	94.00	94.00
Aljameel et al. [40]	97.43	97.00	97.00	97.00
Ekong et al. [10]	97.00	97.00	97.00	97.00
DeepPipeNet	98.51	98.43	98.32	98.31

Table 11 Performance comparison of DeepPipeNet with existing studies on oil pipeline accidents dataset

Study	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Awuku et al. [41]	85.25	85.21	85.23	86.22
Lee et al. [42]	95.83	95.85	95.82	95.83
Ren et al. [43]	97.45	97.45	97.44	97.41
DeepPipeNet	98.29	98.19	98.03	98.26

sense subtle operational anomalies (e.g., mechanical fatigue, excavation damage) that are not captured by ordinary ML models.

Lee et al. [42] came up with a pipeline anomaly detector system via CNN, which was applied on underground pipelines with 95 percent accuracy. Their model is restricted by the use of accelerators and fixed-depth CNNs which restrict its ability to be flexible across pipeline settings (e.g., different pipe diameter, noisy conditions). DeepPipeNet overcomes it through multi-model feature fusion (VGG16, ResNet50, and DenseNet121) and attention mechanisms and improves the accuracy by 2.46 percent enhancement.

Ren et al. [43] suggested an adversarial temporal convolutional network (ATCN) to predict oil spill drifts with 97.45 percent accuracy. Their system is very effective when it comes to temporal sea-surface motions, but does not handle spatial feature extraction that is required in identifying pipeline accidents (e.g., corrosion, weld failure). The hybrid architecture of DeepPipeNet is a solution to this flaw, as it raises the accuracy as well as by 0.84% and has been shown to be generalizable to a variety of pipeline conditions.

DeepPipeNet has shown and continues to show excellent performance in two domain-specific and comprehensive datasets of the most common and stressful failures in oil and gas pipelines. The targeted validation plan has allowed us to develop a strong standard

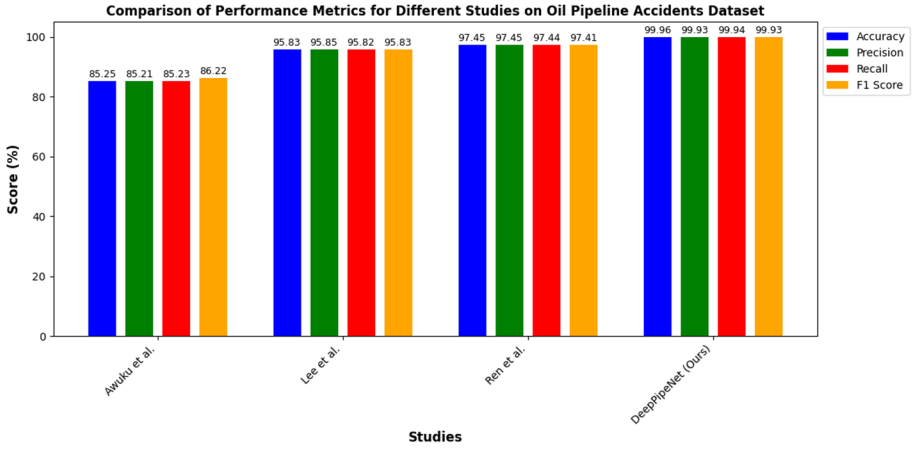


Fig. 38 Performance comparison of existing studies with proposed DeepPipeNet on oil pipeline accidents dataset

of AI-based pipeline monitoring, with state-of-the-art results regarding the detection of severity of corrosion and the causes of accidents. Based on this solid foundation, a number of interesting evolution directions become available to expand the capacities of the framework. Further developments can also encompass information on different geographical areas and pipes to promote overall generalization in the global infrastructure. Secondly, it would further strengthen the ability of the system to monitor the system in terms of its inherent monitoring capabilities through the extension of the anomaly taxonomy to include rare events.

The existing multi-model ensemble model achieves the highest level of accuracy using the complementary features representations of VGG16, resnet50 and densenet121. This design allows the highest degree of detection reliability in safety–critical designs. To optimize the system with computational constraints, we have singled out several optimization paths such as lightweight architecture replacement, model quantization, and knowledge distillation that can easily accommodate high accuracy with minimal computational cost.Besides, based on the powerful detection capabilities we have achieved in this work, our research roadmap would involve several main additions such as adversarial training to enhance the system resilience to cyber-physical threats, semi-supervised learning features that can achieve high accuracy with significantly low computational cost, and modular integration frameworks of other sensor modalities including infrared and acoustic data.

5 Conclusion and future works

The article introduces DeepPipeNet, a hybrid model that is based on deep learning and significantly more effective at detecting anomalies in oil and gas pipes than the existing methods because it fills the gaps left by the existing techniques. The combination of VGG16, ResNet50 and DenseNet121 features along with channel-spatial attention mechanisms make the model achieve a high and solid performance (98.51% on leakage detection, 98.29% on the cause of accidents) with the lowest false positive rate at the lowest possible. The ability to identify hierarchical, residual, and dense spatial patterns in the model enables proper identification of diverse anomalies, including corrosion, leakages, and mechanical errors. SMOTE-based augmentation ensures the strength of the class imbalance, and attention modules choose significant sensor signals (e.g., pressure

drops, CO 2 spikes) against noise. DeepPipeNet enhances state-of-the-art solutions by 2.5–15 percent, that demonstrates the flexibility to adapt to dynamic pipeline conditions and its ability to offer a scalable predictive maintenance solution.Despite success there are limitations of the existing framework. First, its high performance is validated on two specific benchmark datasets. While indicative of strong capability, generalization to the full heterogeneity of global pipeline networks (e.g., subsea, arctic, urban distribution) requires further testing with diverse operational data. Although class imbalance is alleviated by SMOTE, rare anomalies (e.g., sabotage, extreme weather damage) are still underrepresented and could impact generalization. Second, the computational cost of the ensemble architecture, although tolerable in controlled environments, could pose a problem for real-time deployment on resource-limited edge devices. Third, although the attention mechanism improves feature relevance, the interpretability of the model is still incomplete; operators can have no actionable knowledge of why certain anomalies are detected. Lastly, adversarial attacks were not attempted, so robustness to malicious interference remains untested.Future research will address main directions, Enhancing computation efficiency through light-weight architectures (e.g., MobileNet-based feature extractors) and quantization will make it possible for real-time monitoring at the edge. Furthermore, our current attention mechanisms come equip with meaningful visualization of the feature importance, we will build in complementary explainable AI (XAI) tools like Grad-CAM and SHAP's in order to offer multi-faceted interpretability. It will offer operators not only spatial attention maps but also feature contribution analysis, which will give a comprehensive insight into the decisions made by models as well as help them respond to incidents faster and will improve the resistance to cyber-physical attacks through the use of unlabeled data will identify new forms of anomalies, and adversarial training will improve the reliability. Lastly, the combination of DeepPipeNet in pilot projects and pipelines equipped with the IoT will ensure that the implemented effect is justified, creating a relationship between the theoretical performance and the scalability of the industrial. These measures imagine the transformation of DeepPipeNet into a versatile, resilient and explainable pipeline integrity controller, facilitating the safer and more carbon sustainable energy transportation in the world.

Author contributions

Pankaj Verma: Writing—review & editing, Writing—original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. Krishna Gandhi: Writing—review & editing, Visualization, Validation, Formal analysis. Zaid Bin Faheem: Writing—review & editing, Visualization, Validation, Data curation. Muhammad Ayaz: Writing—review & editing, Visualization, Validation, Supervision. El-Hadi M Aggoune: Writing—review & editing, Visualization, Validation, Supervision. AHMAD ELTAIEF: Writing—review & editing, Visualization, Validation.

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Data availability

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent to publication

Not applicable.

Competing interests

The authors declare no competing interests.

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